

Code

Predict the average reaction time through the variables we get from the survey

```
## Warning: package 'leaps' was built under R version 4.3.3

## Warning: package 'olsrr' was built under R version 4.3.3

## Warning: package 'glmnet' was built under R version 4.3.3

## Warning: package 'lmtest' was built under R version 4.3.3

## Warning: package 'zoo' was built under R version 4.3.3

## Warning: package 'nlme' was built under R version 4.3.3

## Warning: package 'car' was built under R version 4.3.3

## Warning: package 'carData' was built under R version 4.3.3

## [1] "First"          "Second"          "Third"           "Class"
## [5] "Age"            "AvgSleepTime"    "LastNightSleep"  "HoursAweak"
## [9] "Fatigue"        "Stress"          "Distraction"     "NoiseLevel"
## [13] "Temperature"    "VideoGamePlay"   "SportsPlay"      "ShootingGames"
## [17] "AvgHoursExercise" "Gamer"           "CaffeinIntake"   "AlcoholIntake"
## [21] "VisualAcuity"   "PrimarilyHand"   "UsePrimaryHand"  "Cautious"
## [25] "InputDevice"    "RefreshRate"     "DeviceOS"        "WiFi"

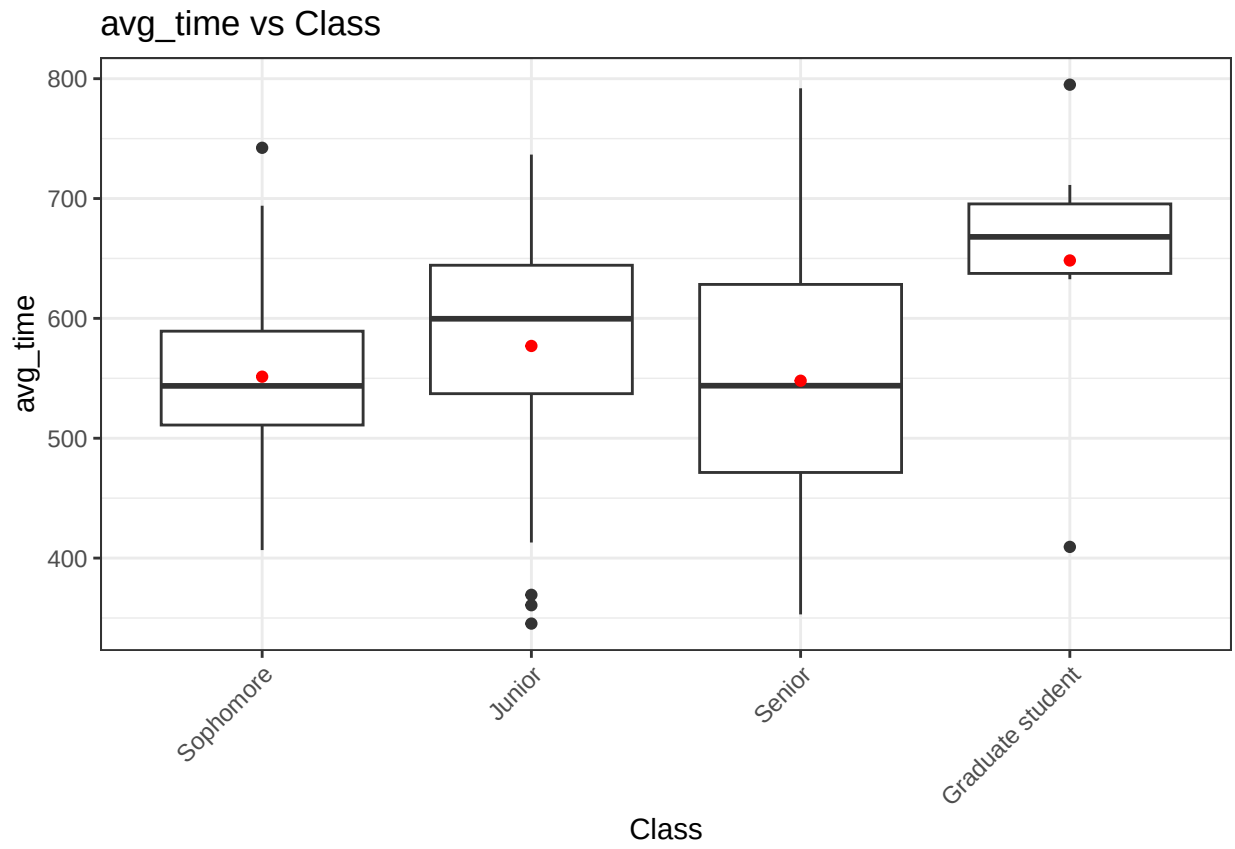
##      First      Second      Third      Class
## Min.   :325.0  Min.   :338.0  Min.   :341.0  Sophomore    :45
## 1st Qu.:520.0  1st Qu.:498.0  1st Qu.:490.0  Junior       :55
## Median :593.0  Median :561.0  Median :542.0  Senior       :22
## Mean   :592.5  Mean   :560.7  Mean   :547.7  Graduate student: 7
## 3rd Qu.:663.0  3rd Qu.:633.0  3rd Qu.:613.0
## Max.   :934.0  Max.   :833.0  Max.   :783.0
##      Age      AvgSleepTime  LastNightSleep  HoursAweak
## Min.   :18.0  Min.   : 5.000  Min.   : 4.000  Min.   : 0.000
## 1st Qu.:20.0  1st Qu.: 6.500  1st Qu.: 6.000  1st Qu.: 4.000
## Median :20.0  Median : 7.000  Median : 7.000  Median : 6.000
## Mean   :20.9  Mean   : 7.236  Mean   : 7.264  Mean   : 7.322
## 3rd Qu.:21.0  3rd Qu.: 8.000  3rd Qu.: 8.000  3rd Qu.:10.000
## Max.   :26.0  Max.   :10.000  Max.   :11.000  Max.   :18.000
##      Fatigue      Stress      Distraction      NoiseLevel
```

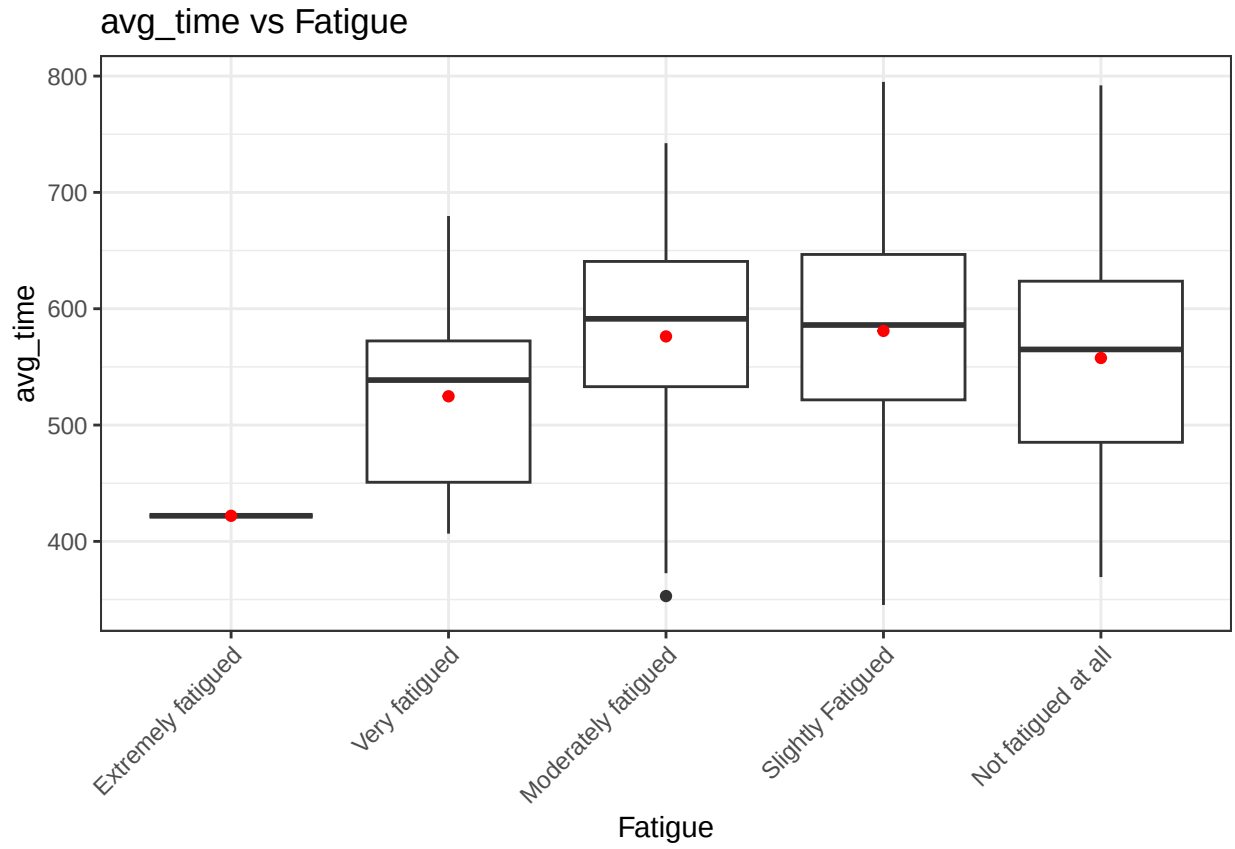
```

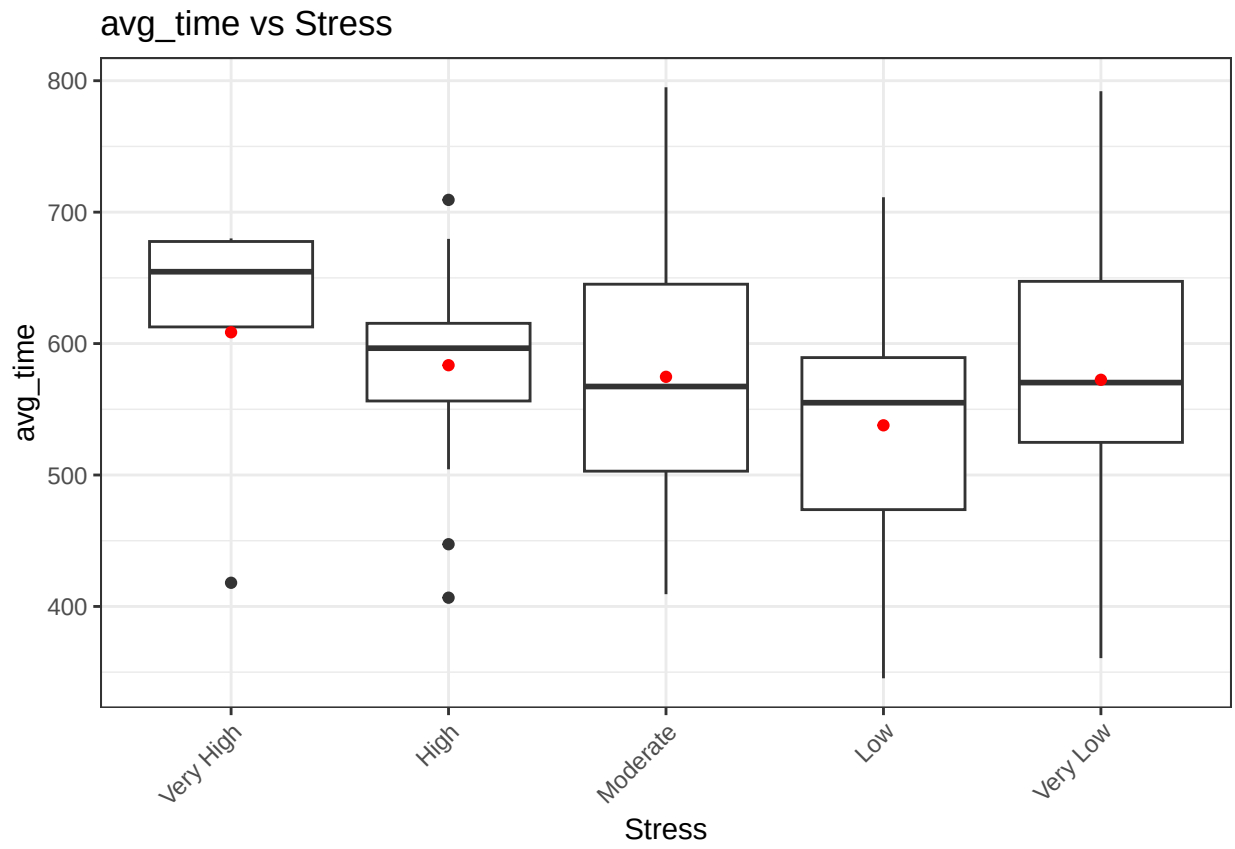
## Extremely fatigued : 1 Very High: 5 Length:129 Min. : 1.000
## Very fatigued :16 High :20 Class :character 1st Qu.: 2.000
## Moderately fatigued:44 Moderate :46 Mode :character Median : 3.000
## Slightly Fatigued :45 Low :35 Mean : 3.287
## Not fatigued at all:23 Very Low :23 3rd Qu.: 5.000
## Max. :10.000
## Temperature VideoGamePlay SportsPlay
## Very Warm: 2 Daily :11 Daily : 5
## Warm :16 Several times a week :36 Several times a week :24
## Neutral :98 Once a week : 9 Once a week :11
## Cold :13 Several times a month:24 Several times a month:24
## Rarely :38 Rarely :50
## Never :11 Never :15
## ShootingGames AvgHoursExercise Gamer CaffeinIntake
## None :19 Min. : 0.000 Length:129 Length:129
## Beginner :54 1st Qu.: 2.000 Class :character Class :character
## Intermediate:35 Median : 4.000 Mode :character Mode :character
## Advanced :18 Mean : 4.097
## Expert : 3 3rd Qu.: 6.000
## Max. :12.000
## VisualAcuity PrimarilyHand UsePrimaryHand
## Length:129 Right hand :119 Yes:122
## Class :character Left hand : 6 No : 7
## Mode :character Ambidextrous (both hands equally): 4
##
##
## Cautious InputDevice RefreshRate
## Extremely cautious : 3 Game controller: 1 High :29
## Very cautious :32 Mouse :47 Mid-Range:52
## Moderately cautious:68 Keyboard : 2 Standard :48
## Slightly cautious :21 Trackpad :69
## Not cautious at all: 5 Touch screen :10
##
## DeviceOS WiFi avg_time
## Length:129 Unstable: 2 Min. :345.3
## Class :character Stable :127 1st Qu.:511.0
## Mode :character Median :574.0
## Mean :567.0
## 3rd Qu.:635.0
## Max. :795.0

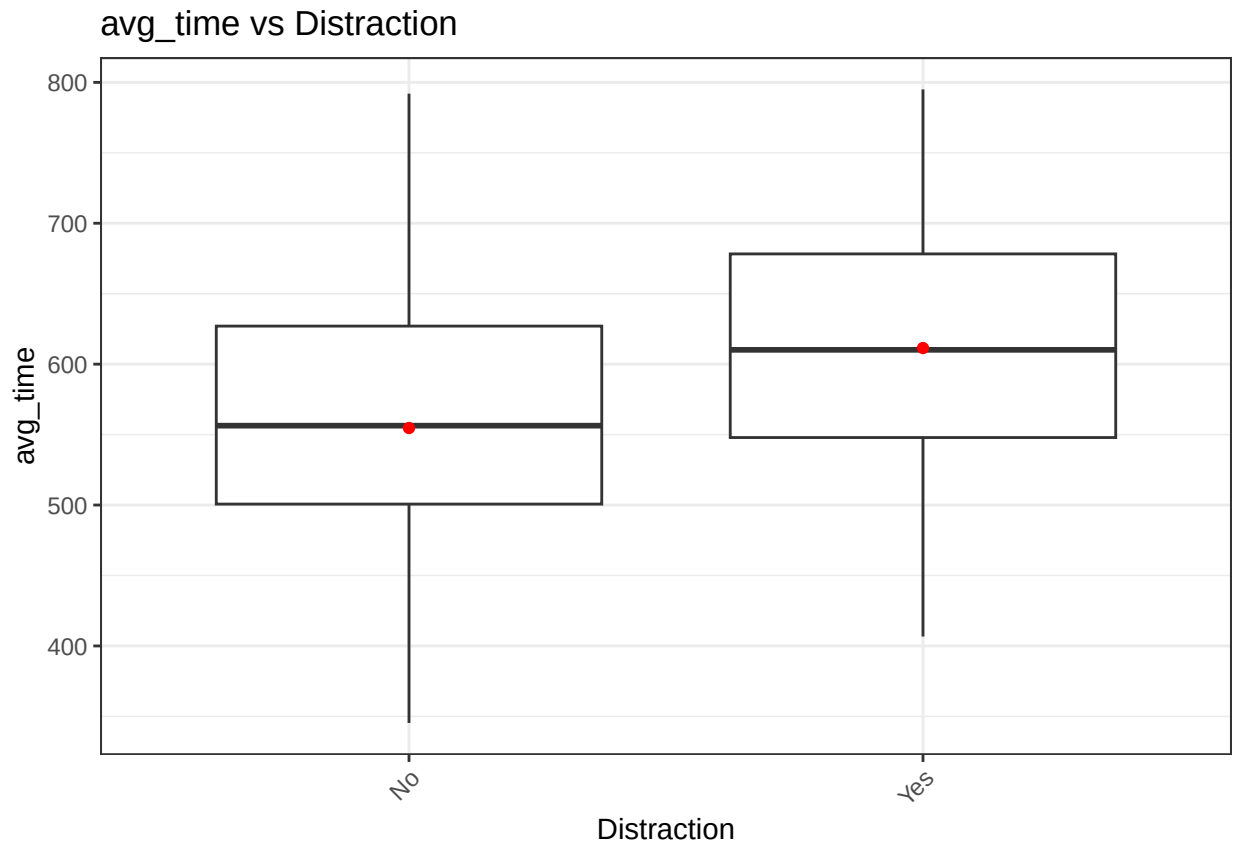
## Warning: 'aes_string()' was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with 'aes()'.
## i See also 'vignette("ggplot2-in-packages")' for more information.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

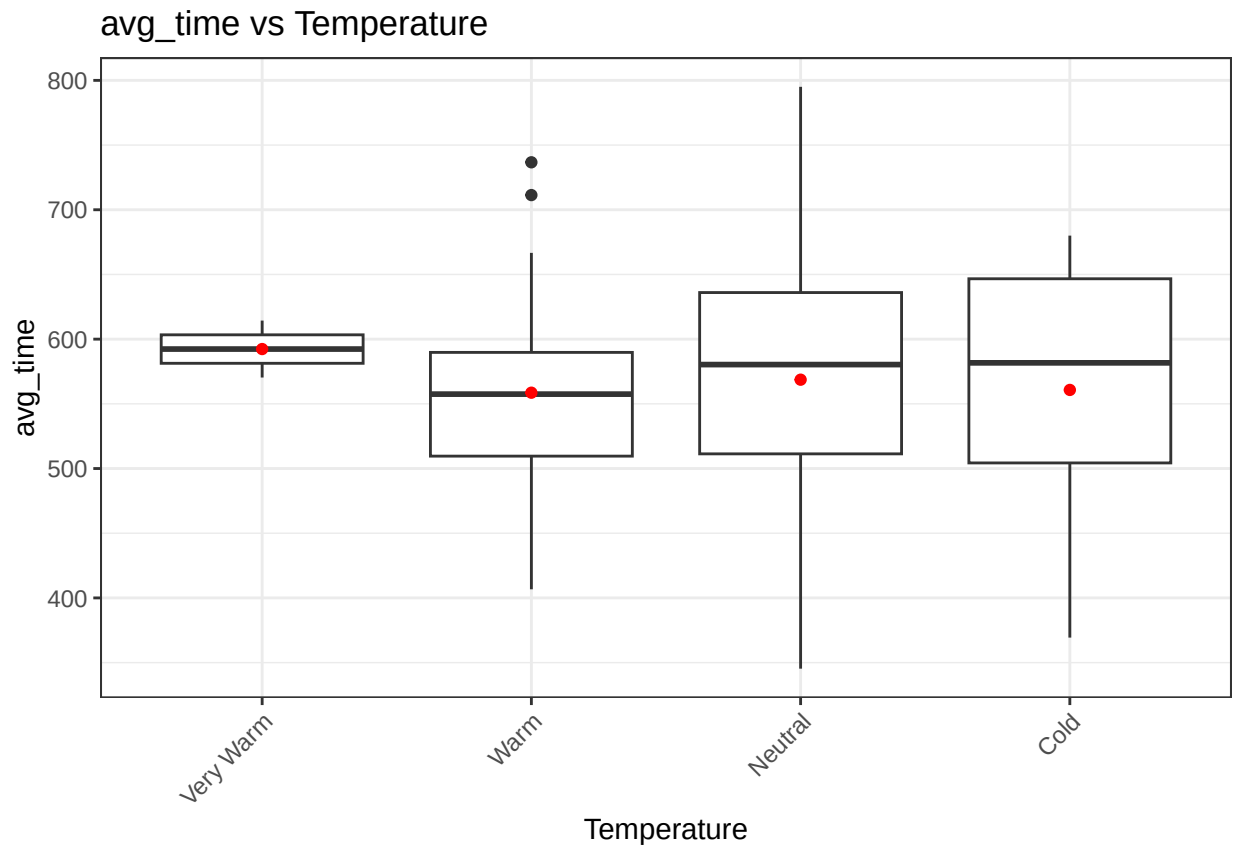
```

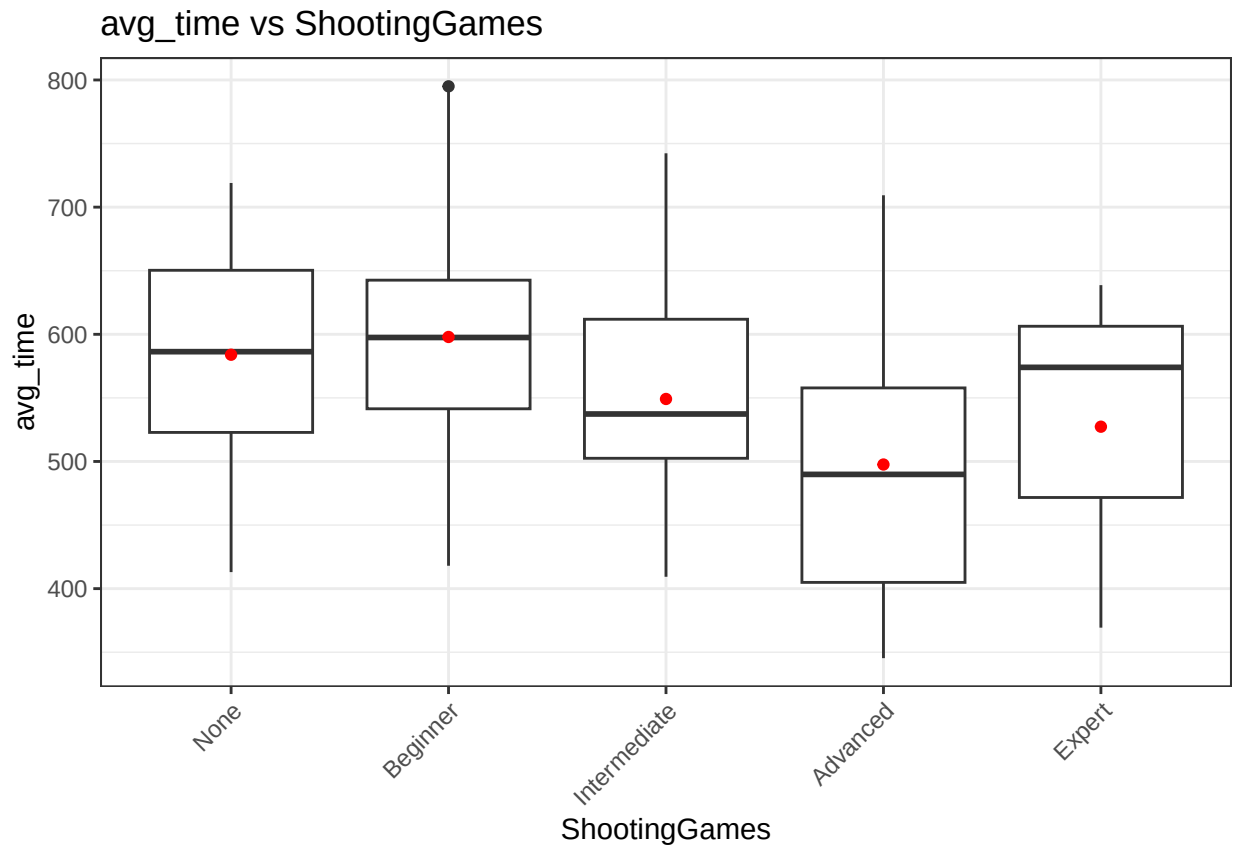


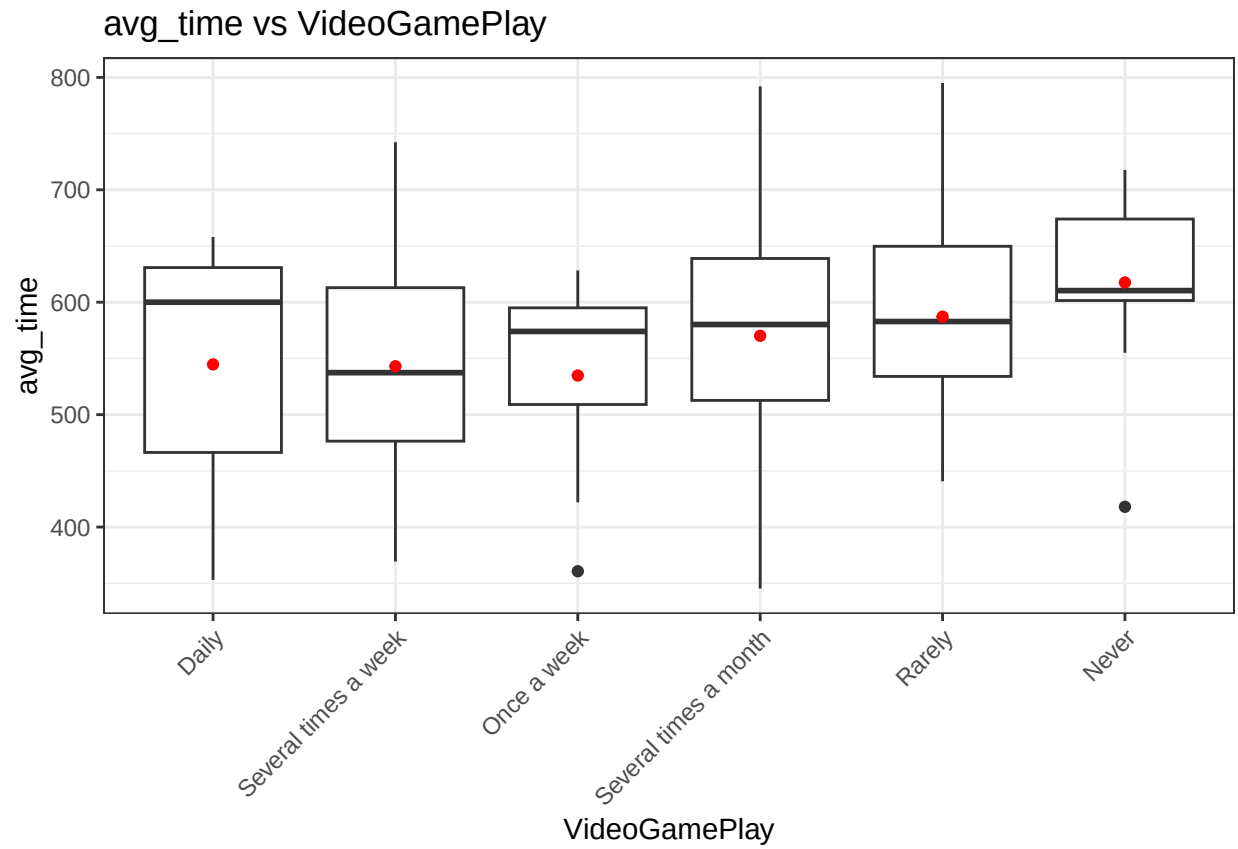


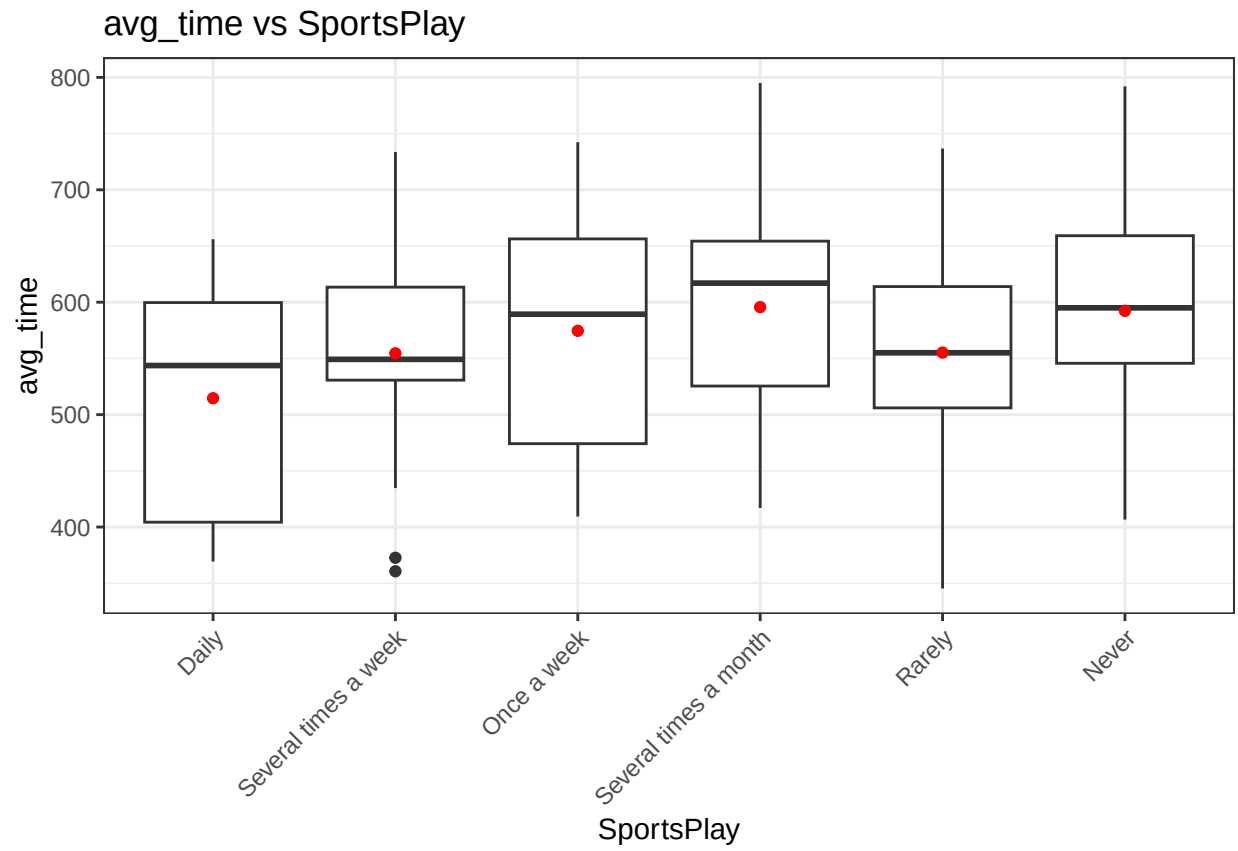


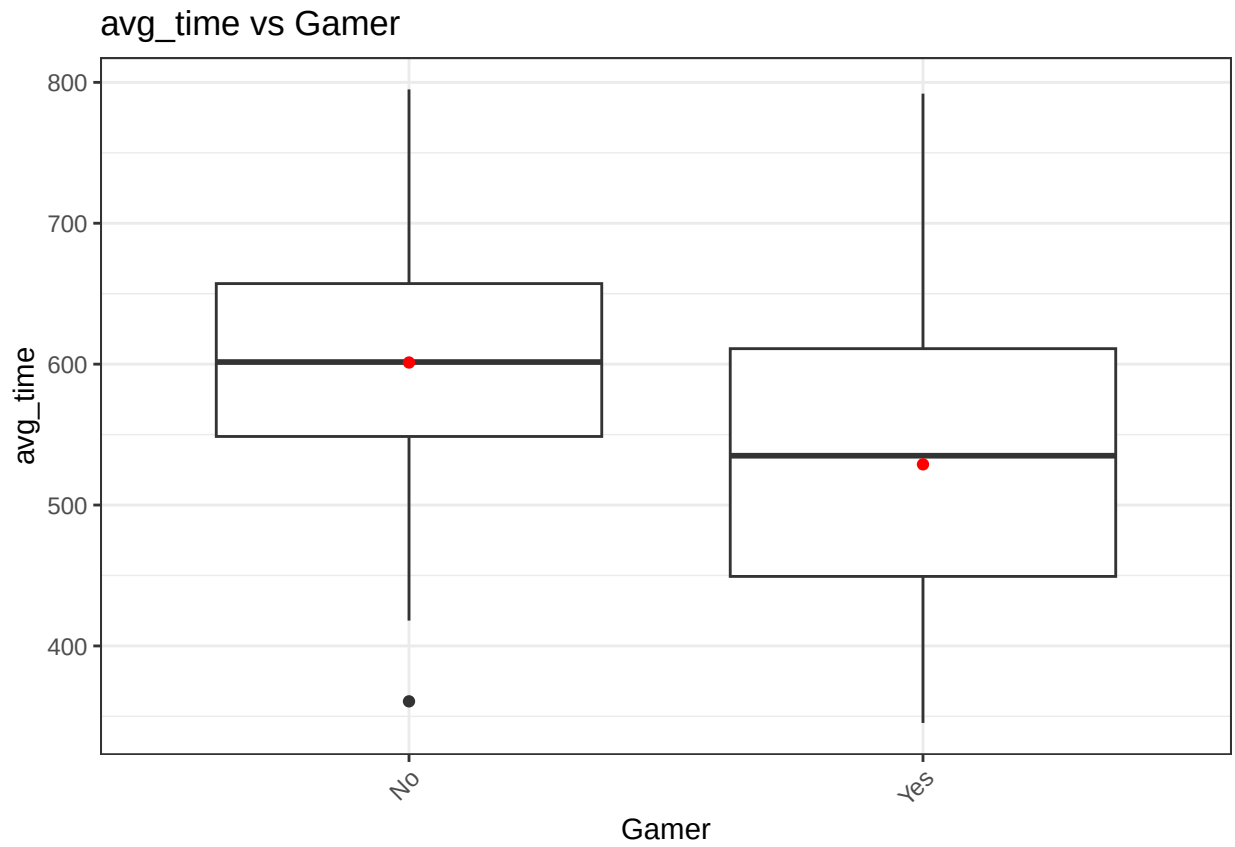


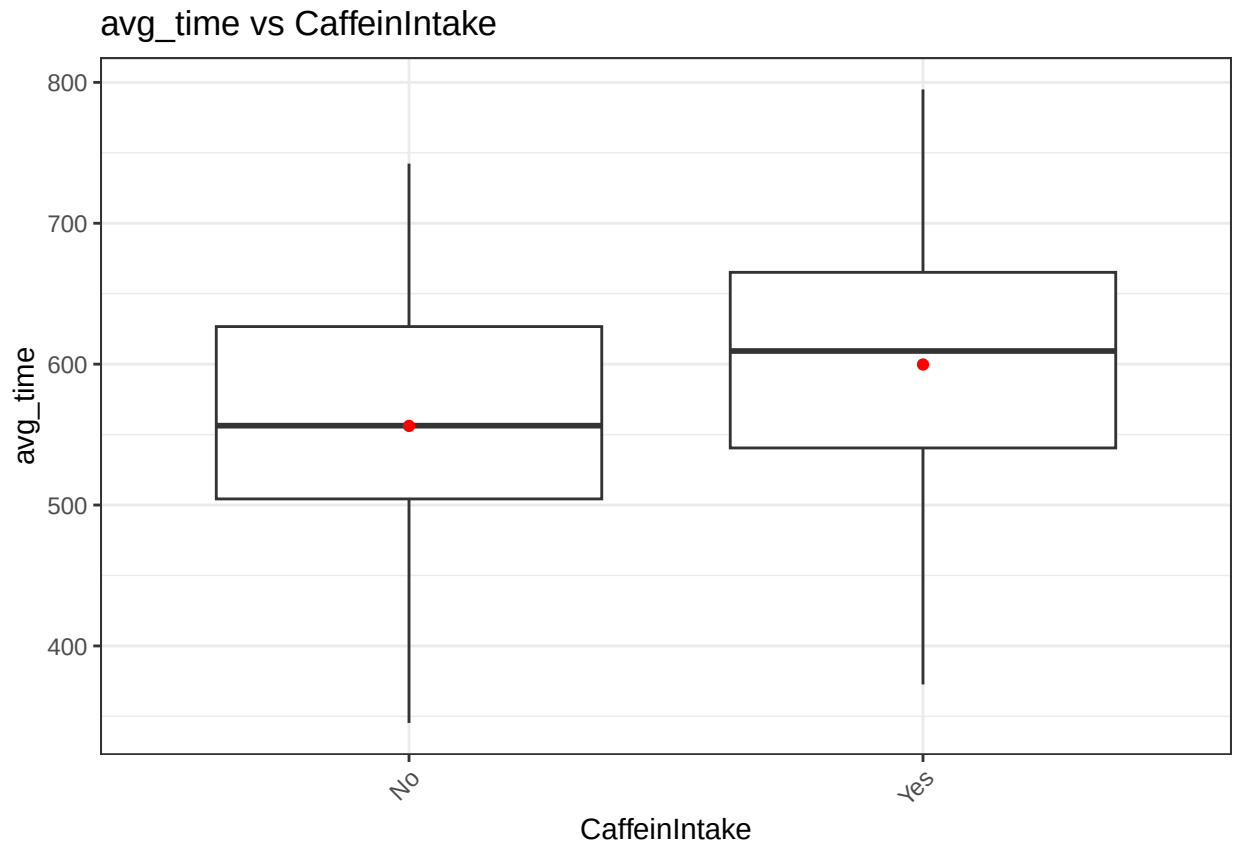


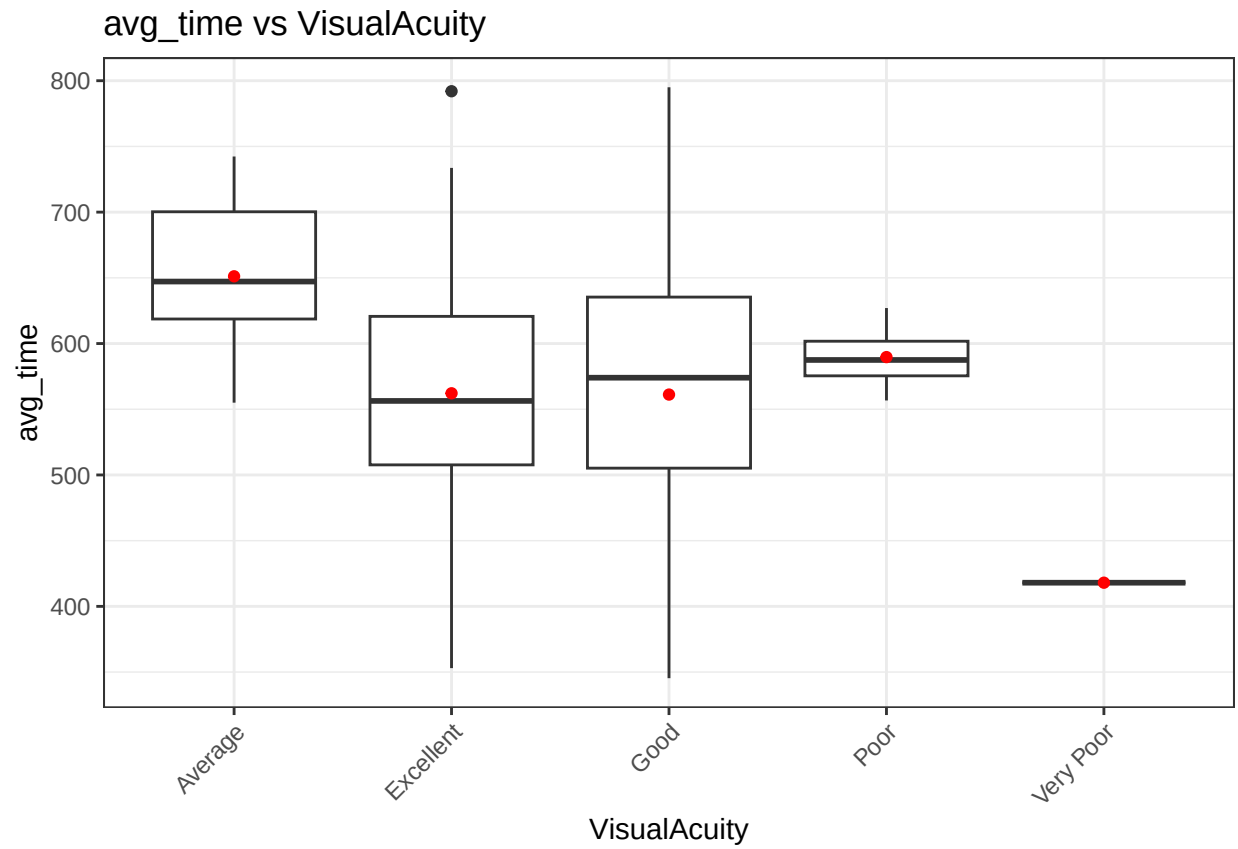


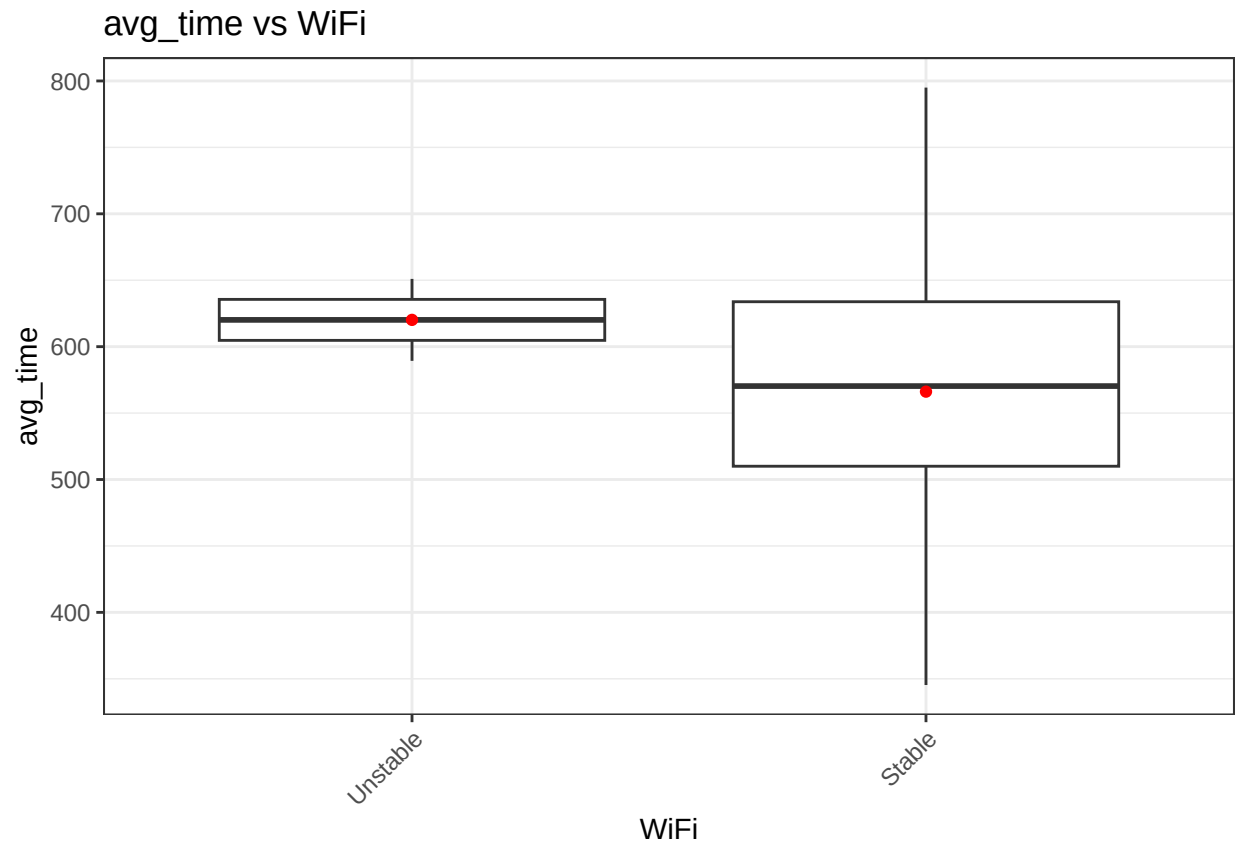


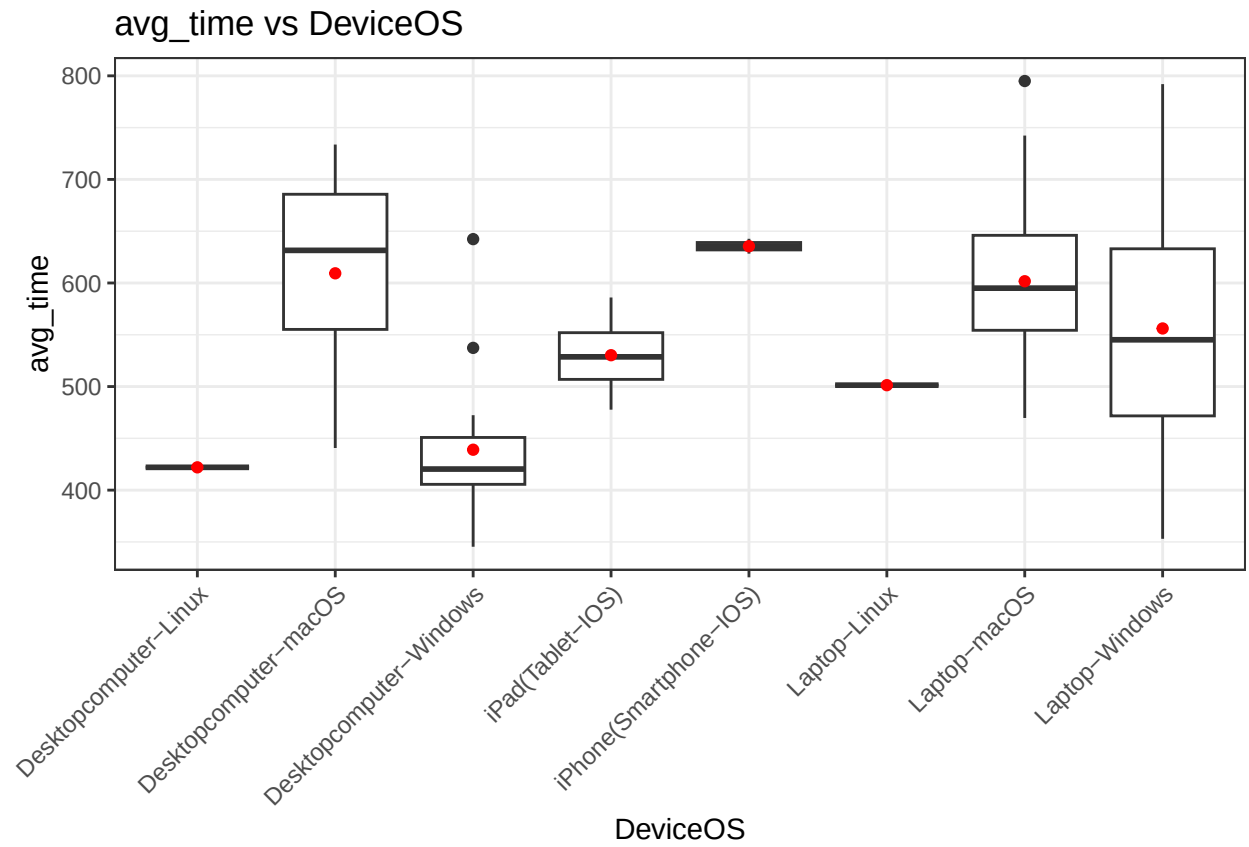


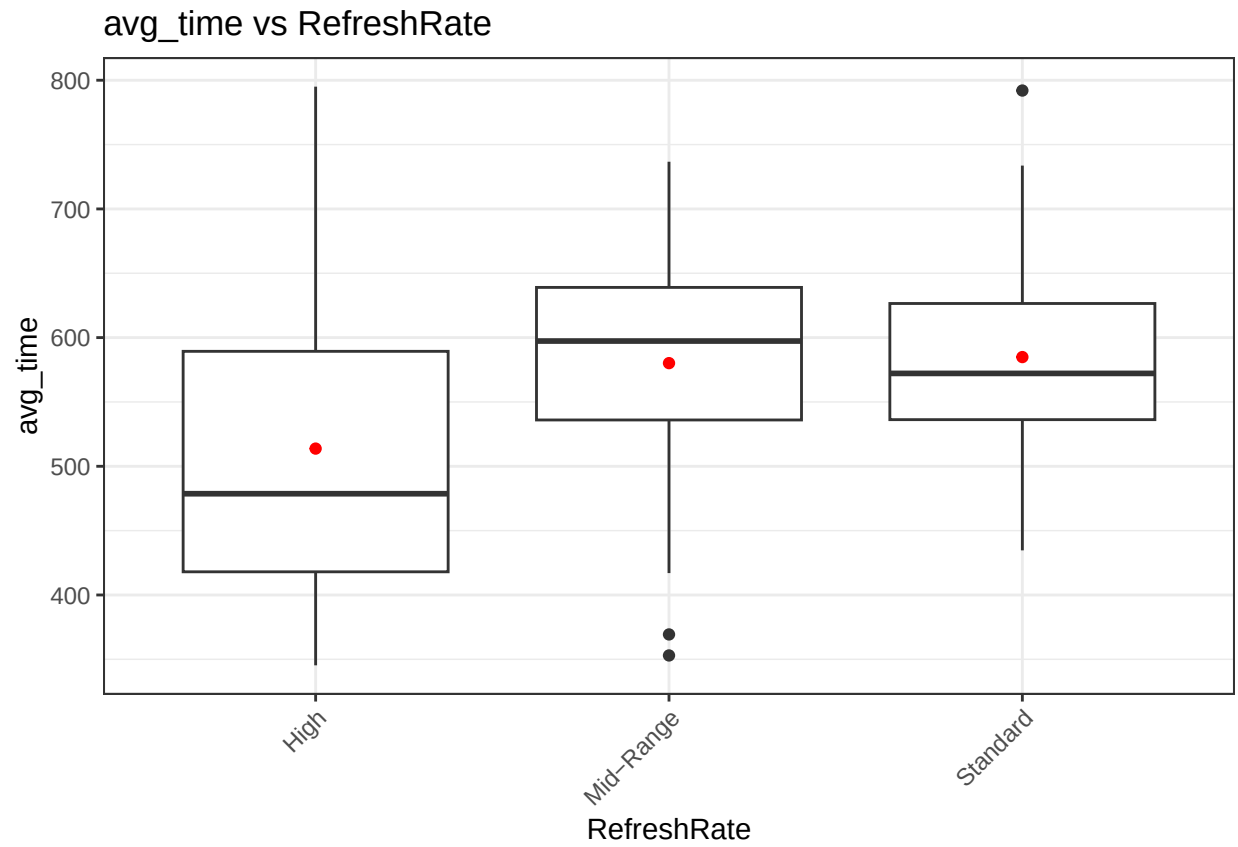


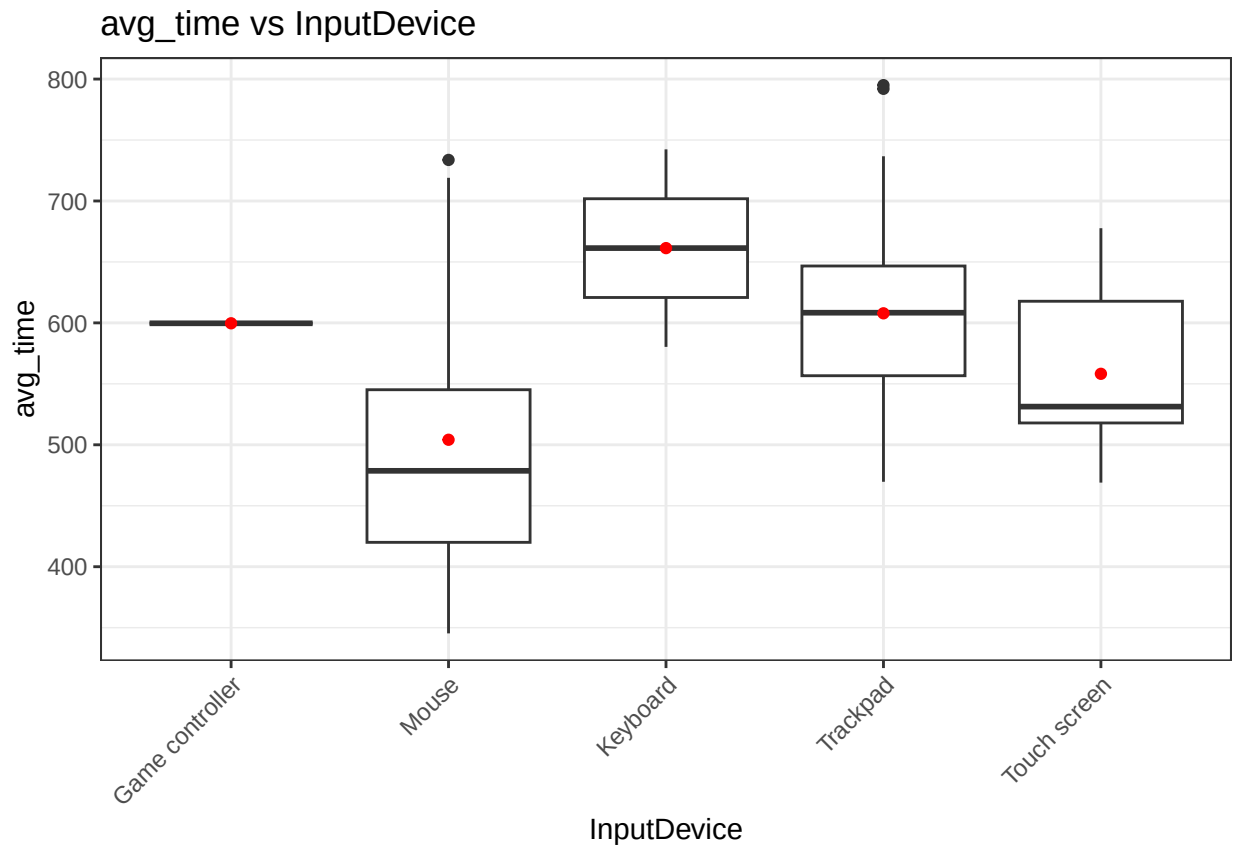


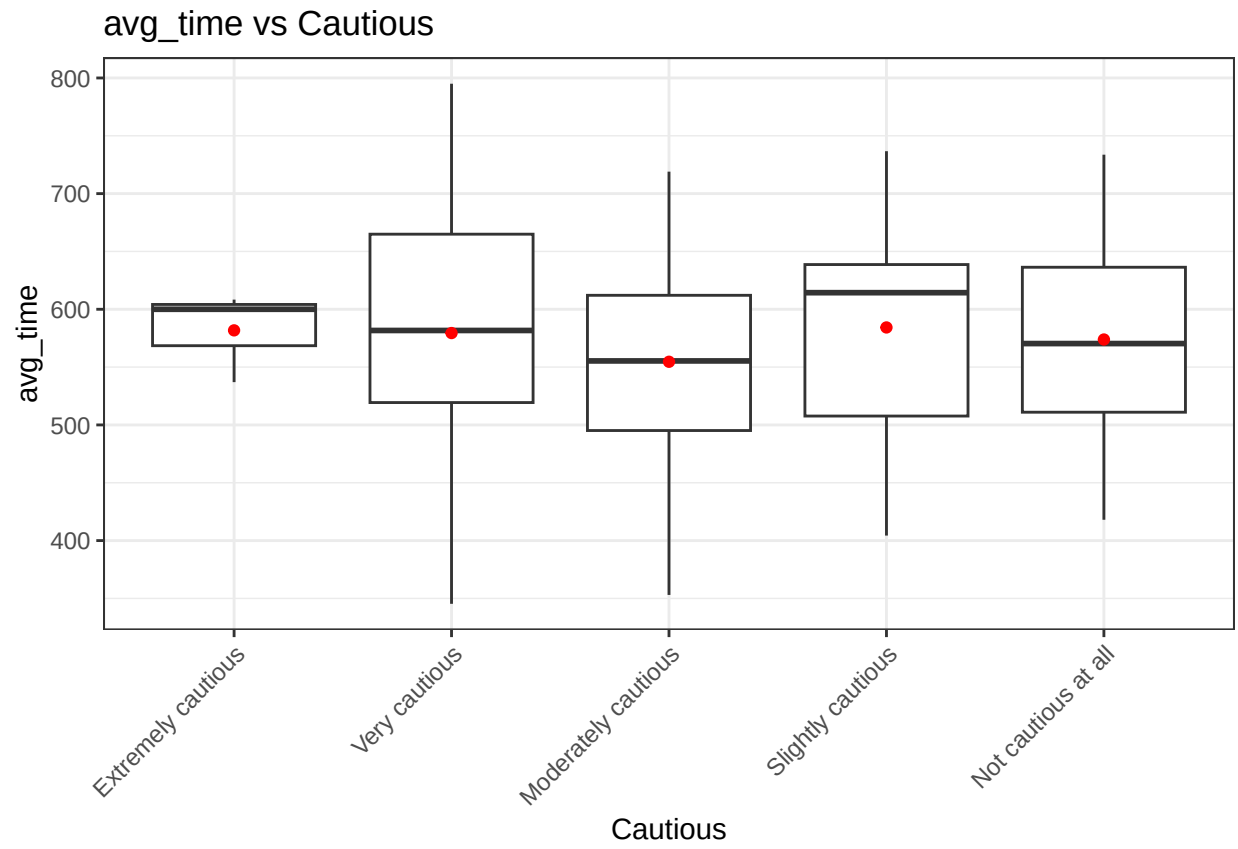


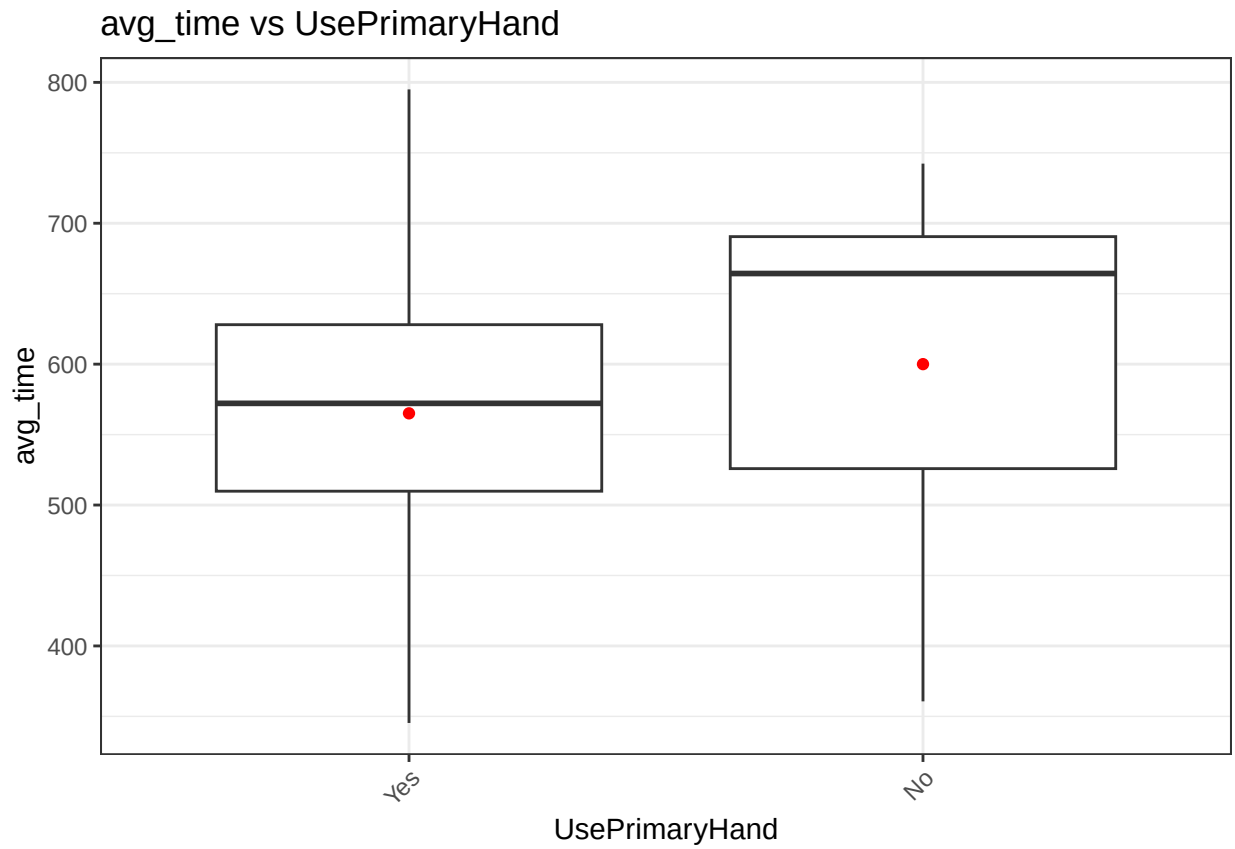


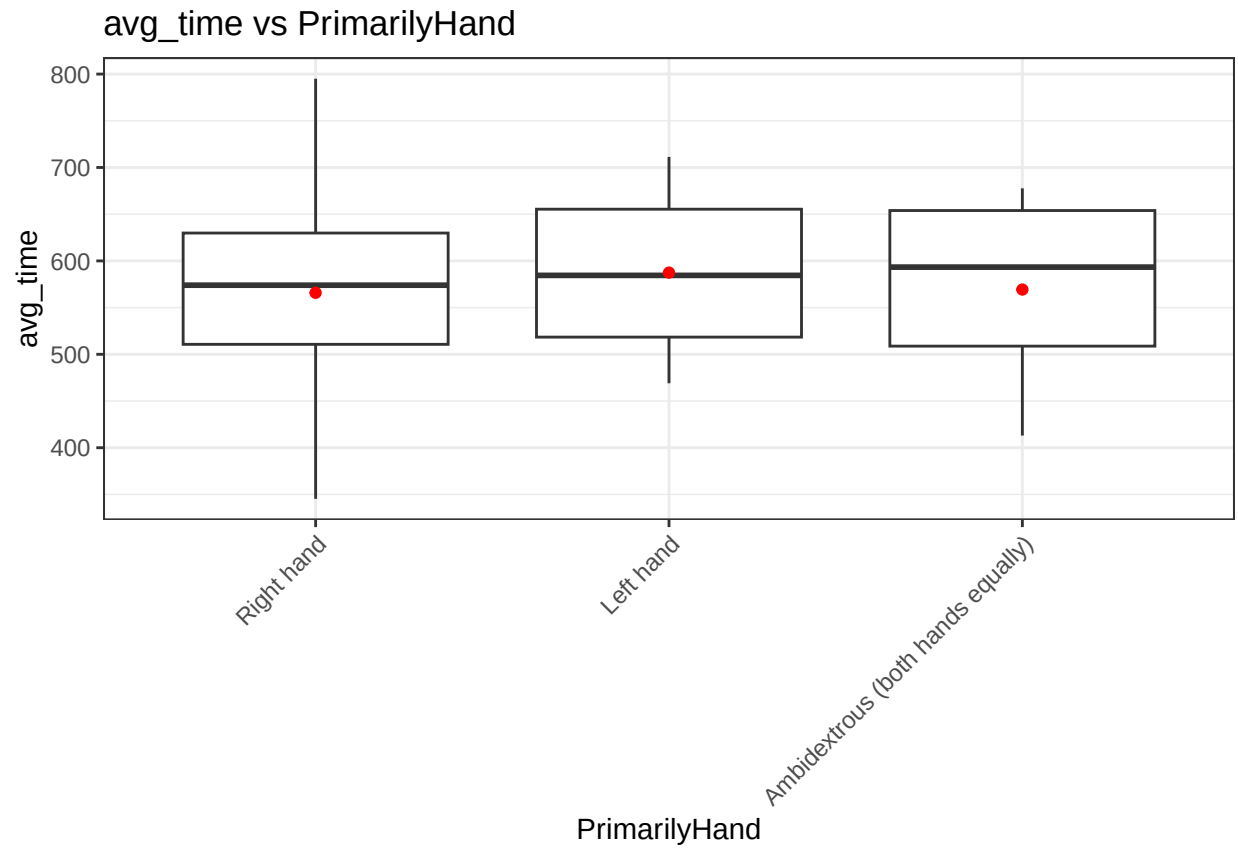


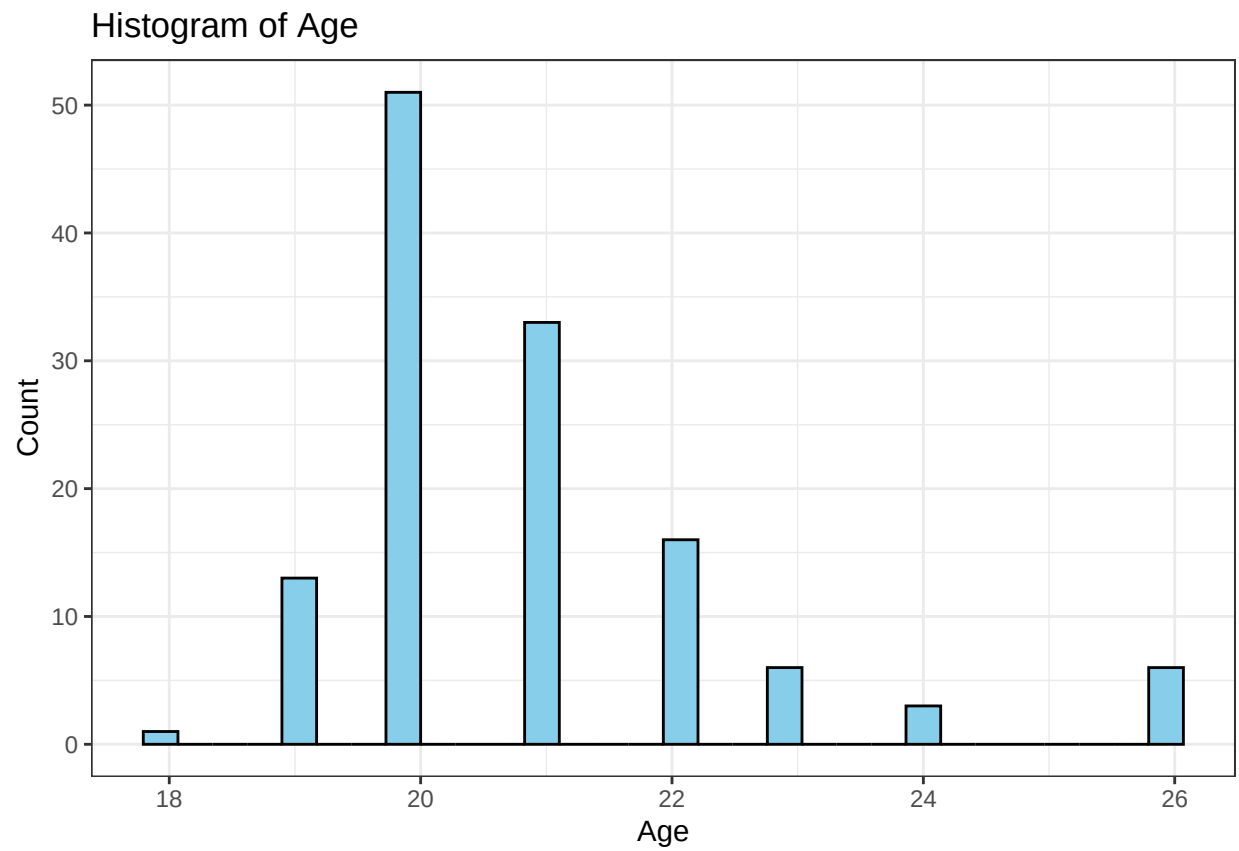


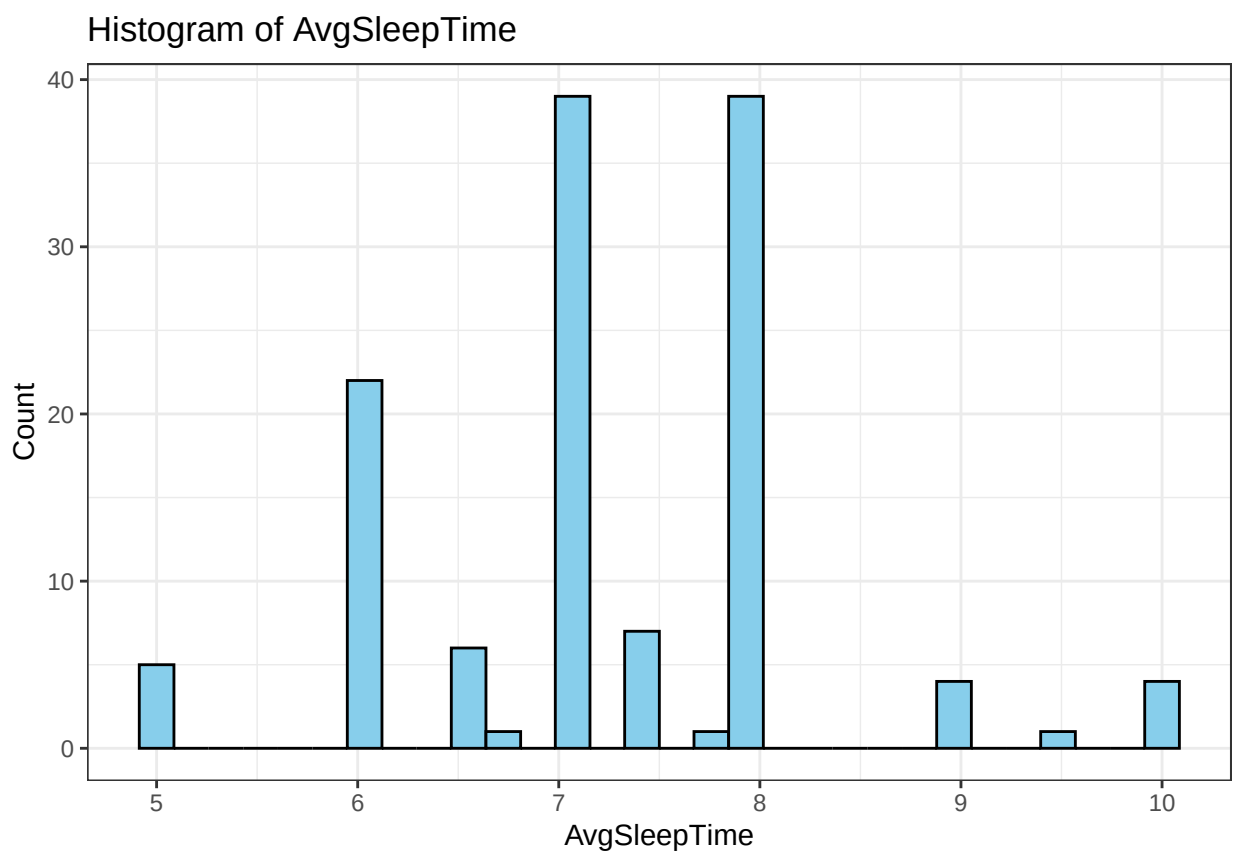


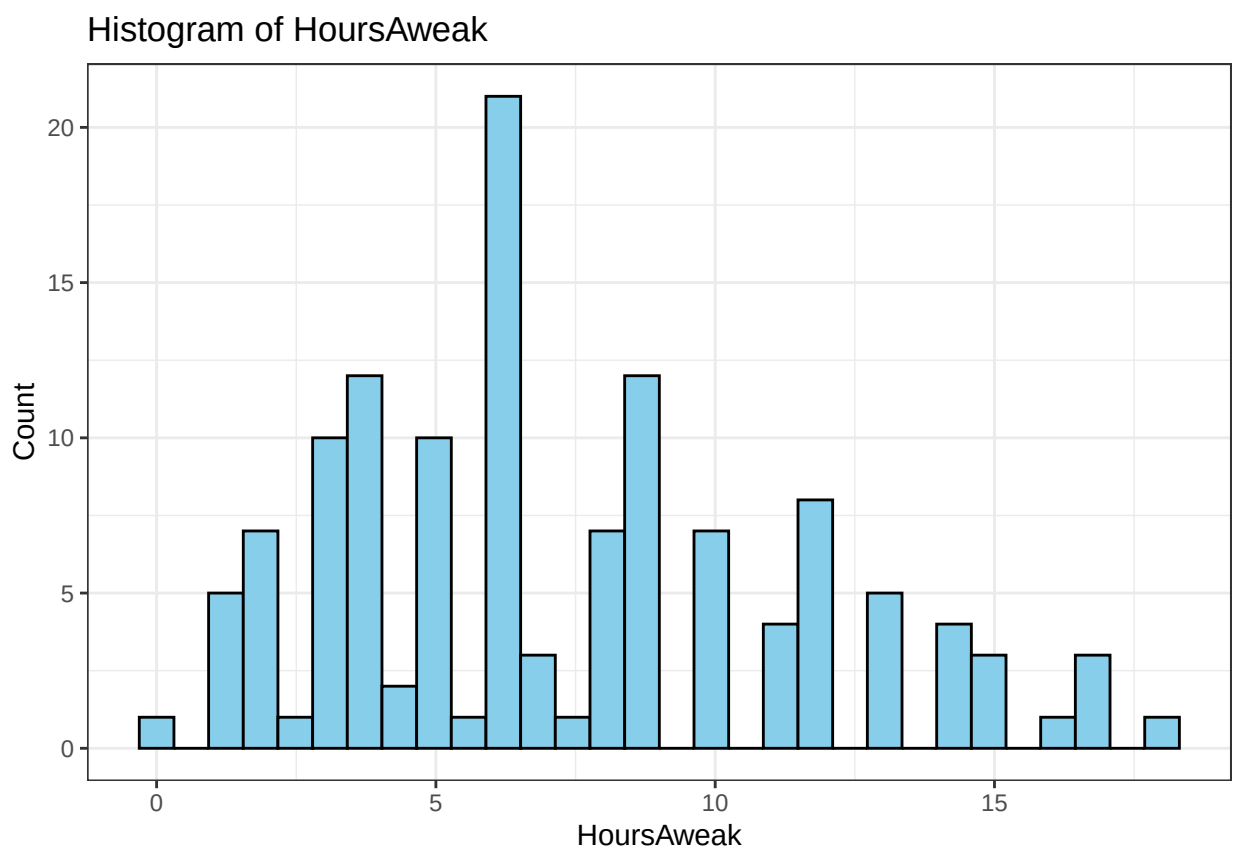


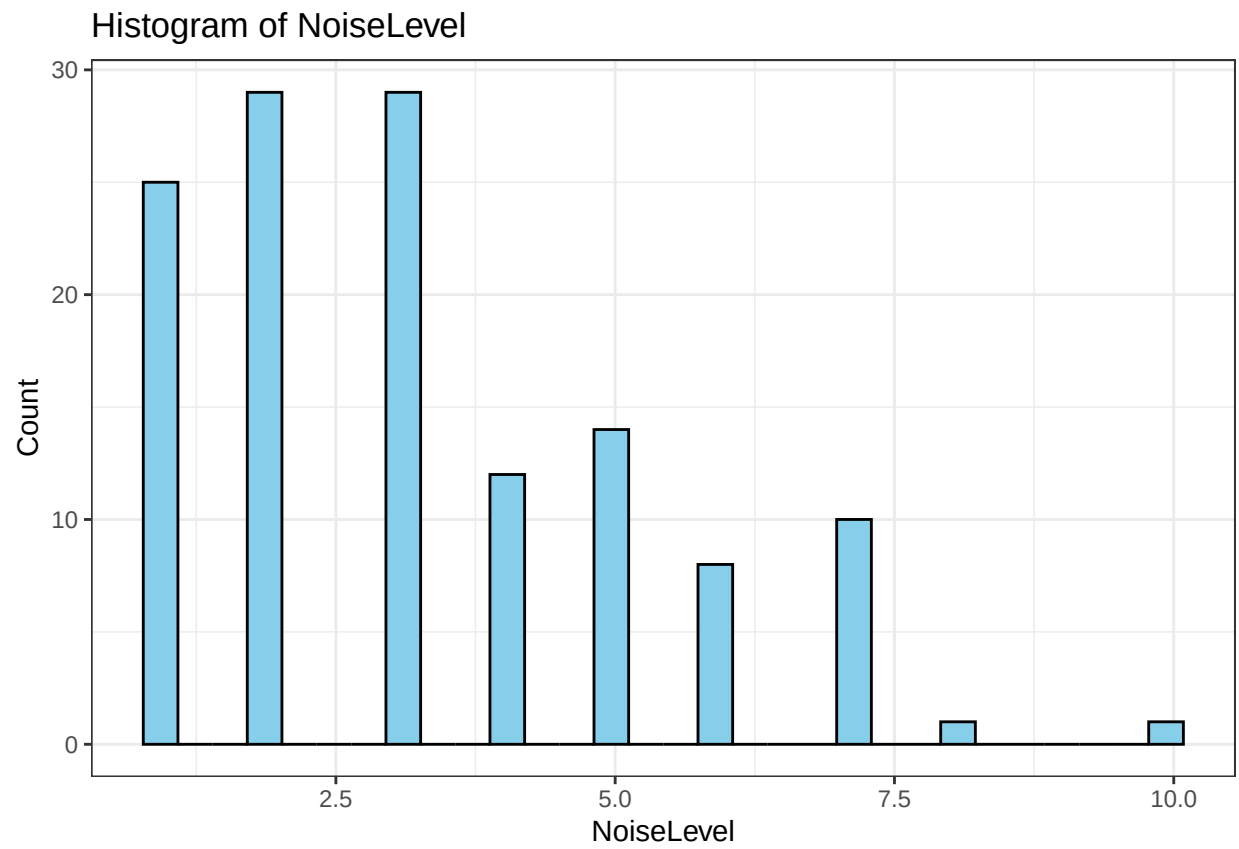


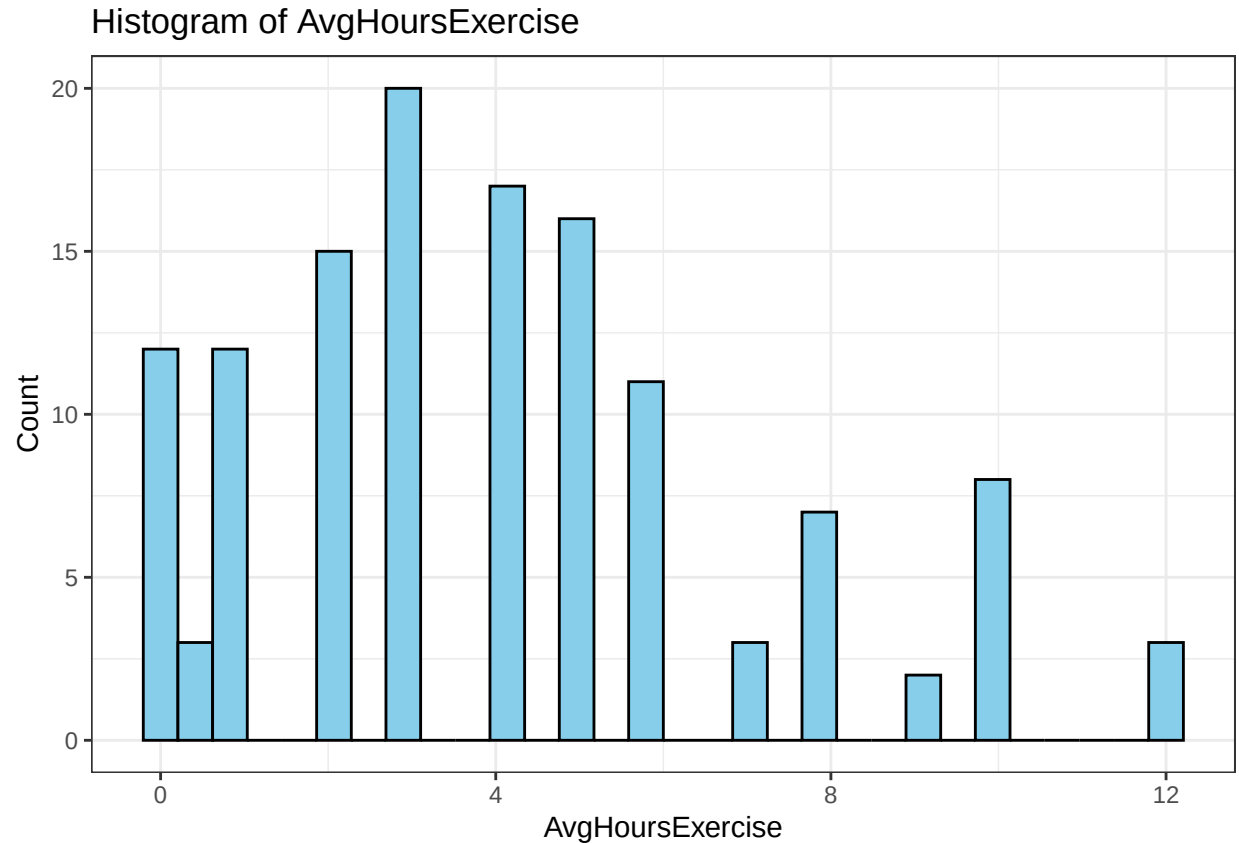












```
##
## Call:
## lm(formula = avg_time ~ ., data = Survey_cleaned)
##
## Residuals:
```

##	Min	1Q	Median	3Q	Max
##	-119.584	-31.677	-1.699	23.259	158.041

```
##
## Coefficients:
```

##	Estimate	Std. Error	t value
## (Intercept)	316.7155	214.0247	1.480
## ClassJunior	36.3028	19.3107	1.880
## ClassSenior	5.9146	27.0350	0.219
## ClassGraduate student	120.6718	55.3700	2.179
## Age	3.0620	8.1732	0.375
## AvgSleepTime	15.7785	8.3699	1.885
## LastNightSleep	1.7912	5.3886	0.332
## HoursAweek	0.5274	2.0285	0.260
## DistractionYes	28.4336	20.0834	1.416
## TemperatureWarm	25.1289	64.2955	0.391
## TemperatureNeutral	46.3851	60.6552	0.765
## TemperatureCold	65.9307	63.1520	1.044
## GamerYes	15.4718	18.0549	0.857
## CaffeinIntakeYes	40.1718	17.9718	2.235
## VisualAcuityExcellent	-58.0343	29.0626	-1.997
## VisualAcuityGood	-79.1387	29.9986	-2.638

## VisualAcuityPoor	-72.0108	50.5460	-1.425
## VisualAcuityVery Poor	-146.9141	97.1368	-1.512
## PrimarilyHandLeft hand	-54.6740	44.0925	-1.240
## PrimarilyHandAmbidextrous (both hands equally)	34.9272	40.1681	0.870
## UsePrimaryHandNo	65.8135	40.6695	1.618
## CautiousVery cautious	3.0895	45.3541	0.068
## CautiousModerately cautious	-6.5308	46.0642	-0.142
## CautiousSlightly cautious	9.0990	48.4746	0.188
## CautiousNot cautious at all	38.3024	64.5652	0.593
## InputDeviceMouse	-45.1659	78.2158	-0.577
## InputDeviceKeyboard	77.1231	96.5755	0.799
## InputDeviceTrackpad	37.3924	77.5345	0.482
## InputDeviceTouch screen	18.3343	85.9597	0.213
## RefreshRateMid-Range	30.3461	24.3793	1.245
## RefreshRateStandard	59.9600	22.9308	2.615
## DeviceiPad(Tablet	-42.5427	64.8587	-0.656
## DeviceiPhone(Smartphone	81.0134	83.1190	0.975
## DeviceLaptop	36.2946	24.0861	1.507
## WiFiStable	-96.2096	58.0745	-1.657
## Fatigue_3Moderately Fatigue	54.4543	26.0192	2.093
## Fatigue_3Low Fatigue	48.2829	26.9053	1.795
## Stress_3Moderately Stress	14.0792	20.5947	0.684
## Stress_3Low Stress	-29.3721	21.5548	-1.363
## Noise_LevelModerately Noise	-35.7234	71.8799	-0.497
## Noise_LevelLow Noise	-19.1215	71.4386	-0.268
## Game_FrequencyLow Frequency	21.8759	15.1069	1.448
## Sports_FrequencyLow Frequency	17.6599	18.2598	0.967
## Shooting_LevelsIntermediate Levels	-7.1078	24.4294	-0.291
## Shooting_LevelsLow Levels	21.0557	22.7596	0.925
## SystemLinux	39.9766	67.0464	0.596
## SystemWindows	-0.8867	19.2729	-0.046
## logAvgHoursExercise	3.1802	12.8470	0.248
##	Pr(> t)		
## (Intercept)	0.1428		
## ClassJunior	0.0637 .		
## ClassSenior	0.8274		
## ClassGraduate student	0.0322 *		
## Age	0.7089		
## AvgSleepTime	0.0630 .		
## LastNightSleep	0.7404		
## HoursAweak	0.7955		
## DistractionYes	0.1607		
## TemperatureWarm	0.6969		
## TemperatureNeutral	0.4467		
## TemperatureCold	0.2996		
## GamerYes	0.3940		
## CaffeinIntakeYes	0.0282 *		
## VisualAcuityExcellent	0.0492 *		
## VisualAcuityGood	0.0100 **		
## VisualAcuityPoor	0.1581		
## VisualAcuityVery Poor	0.1343		
## PrimarilyHandLeft hand	0.2186		
## PrimarilyHandAmbidextrous (both hands equally)	0.3871		
## UsePrimaryHandNo	0.1095		

```

## CautiousVery cautious 0.9459
## CautiousModerately cautious 0.8876
## CautiousSlightly cautious 0.8516
## CautiousNot cautious at all 0.5547
## InputDeviceMouse 0.5652
## InputDeviceKeyboard 0.4269
## InputDeviceTrackpad 0.6309
## InputDeviceTouch screen 0.8316
## RefreshRateMid-Range 0.2168
## RefreshRateStandard 0.0106 *
## DeviceiPad(Tablet 0.5137
## DeviceiPhone(Smartphone 0.3326
## DeviceLaptop 0.1357
## WiFiStable 0.1015
## Fatigue_3Moderately Fatigue 0.0395 *
## Fatigue_3Low Fatigue 0.0765 .
## Stress_3Moderately Stress 0.4962
## Stress_3Low Stress 0.1768
## Noise_LevelModerately Noise 0.6205
## Noise_LevelLow Noise 0.7896
## Game_FrequencyLow Frequency 0.1515
## Sports_FrequencyLow Frequency 0.3364
## Shooting_LevelsIntermediate Levels 0.7718
## Shooting_LevelsLow Levels 0.3576
## SystemLinux 0.5527
## SystemWindows 0.9634
## logAvgHoursExercise 0.8051
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 66.77 on 81 degrees of freedom
## Multiple R-squared:  0.6914, Adjusted R-squared:  0.5123
## F-statistic: 3.861 on 47 and 81 DF,  p-value: 4.833e-08

## Start:  AIC=1119.88
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Temperature + Gamer + CaffeinIntake + VisualAcuity +
##   PrimarilyHand + UsePrimaryHand + Cautious + InputDevice +
##   RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Noise_Level +
##   Game_Frequency + Sports_Frequency + Shooting_Levels + System +
##   logAvgHoursExercise
##
##           Df Sum of Sq  RSS   AIC
## - Cautious      4      6186 367259 1114.1
## - System         2       1980 363052 1116.6
## - Noise_Level    2       5074 366147 1117.7
## - Temperature    3      10845 371918 1117.7
## - logAvgHoursExercise 1        273 361346 1118.0
## - HoursAweak     1        301 361374 1118.0
## - LastNightSleep 1        493 361565 1118.0
## - Age            1        626 361699 1118.1
## - Gamer          1       3273 364346 1119.0
## - Sports_Frequency 1       4170 365243 1119.4
## <none>              361073 1119.9

```

```

## - PrimarilyHand      2      11526 372599 1119.9
## - Shooting_Levels    2      11896 372969 1120.1
## - Distraction        1       8935 370008 1121.0
## - Game_Frequency     1       9347 370420 1121.2
## - Device             3      22755 383828 1121.8
## - UsePrimaryHand     1      11674 372746 1122.0
## - WiFi               1      12234 373307 1122.2
## - Fatigue_3          2      19740 380813 1122.7
## - AvgSleepTime       1      15842 376915 1123.4
## - Stress_3           2      23328 384401 1124.0
## - VisualAcuity       4      37145 398218 1124.5
## - CaffeinIntake      1      22272 383345 1125.6
## - Class              3      40144 401217 1127.5
## - RefreshRate        2      34921 395994 1127.8
## - InputDevice        4      73291 434364 1135.7
##
## Step: AIC=1114.07
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Temperature + Gamer + CaffeinIntake + VisualAcuity +
##   PrimarilyHand + UsePrimaryHand + InputDevice + RefreshRate +
##   Device + WiFi + Fatigue_3 + Stress_3 + Noise_Level + Game_Frequency +
##   Sports_Frequency + Shooting_Levels + System + logAvgHoursExercise
##
##           Df Sum of Sq   RSS   AIC
## - System      2      3458 370717 1111.3
## - Temperature  3     11208 378467 1112.0
## - logAvgHoursExercise 1         0 367259 1112.1
## - Age          1       450 367708 1112.2
## - LastNightSleep 1       519 367778 1112.2
## - HoursAweak   1       611 367869 1112.3
## - Noise_Level  2     8204 375463 1112.9
## - Gamer        1     3617 370875 1113.3
## - Sports_Frequency 1     4109 371368 1113.5
## - PrimarilyHand 2    10345 377604 1113.7
## <none>                367259 1114.1
## - Shooting_Levels  2    14422 381681 1115.0
## - Distraction     1    10066 377324 1115.6
## - Device          3    22527 389786 1115.8
## - Game_Frequency  1    10895 378153 1115.8
## - WiFi            1    12477 379736 1116.4
## - UsePrimaryHand  1    15475 382733 1117.4
## - Fatigue_3       2    23869 391128 1118.2
## - Stress_3        2    23955 391213 1118.2
## - VisualAcuity    4    39758 407017 1119.3
## - CaffeinIntake   1    22391 389650 1119.7
## + Cautious        4       6186 361073 1119.9
## - AvgSleepTime    1    26198 393457 1121.0
## - Class           3    40028 407286 1121.4
## - RefreshRate     2    36741 404000 1122.4
## - InputDevice     4    74330 441589 1129.8
##
## Step: AIC=1111.28
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Temperature + Gamer + CaffeinIntake + VisualAcuity +

```

```

##      PrimarilyHand + UsePrimaryHand + InputDevice + RefreshRate +
##      Device + WiFi + Fatigue_3 + Stress_3 + Noise_Level + Game_Frequency +
##      Sports_Frequency + Shooting_Levels + logAvgHoursExercise
##
##      Df Sum of Sq    RSS    AIC
## - logAvgHoursExercise 1      18 370734 1109.3
## - HoursAweak          1      91 370807 1109.3
## - Temperature         3    11934 382651 1109.4
## - Noise_Level         2     6813 377529 1109.6
## - Age                 1     1077 371794 1109.7
## - LastNightSleep      1     1080 371797 1109.7
## - Gamer               1     4380 375097 1110.8
## - PrimarilyHand       2    10668 381384 1110.9
## - Sports_Frequency    1     5524 376240 1111.2
## <none>                370717 1111.3
## - Distraction         1     9422 380139 1112.5
## - Shooting_Levels     2    15720 386437 1112.6
## - Game_Frequency      1    10570 381286 1112.9
## - Device              3    23286 394002 1113.1
## - WiFi                1    13732 384449 1114.0
## + System              2     3458 367259 1114.1
## - Fatigue_3           2    21150 391867 1114.4
## - UsePrimaryHand      1    16142 386858 1114.8
## - Stress_3            2    26900 397616 1116.3
## + Cautious            4     7664 363052 1116.6
## - VisualAcuity        4    40603 411320 1116.7
## - CaffeinIntake       1    23349 394066 1117.2
## - AvgSleepTime        1    23564 394280 1117.2
## - Class               3    37904 408621 1117.8
## - RefreshRate         2    33438 404154 1118.4
## - InputDevice         4   105407 476123 1135.6
##
## Step:  AIC=1109.28
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##      Distraction + Temperature + Gamer + CaffeinIntake + VisualAcuity +
##      PrimarilyHand + UsePrimaryHand + InputDevice + RefreshRate +
##      Device + WiFi + Fatigue_3 + Stress_3 + Noise_Level + Game_Frequency +
##      Sports_Frequency + Shooting_Levels
##
##      Df Sum of Sq    RSS    AIC
## - HoursAweak          1      87 370821 1107.3
## - Temperature         3    11961 382695 1107.4
## - Noise_Level         2     6820 377555 1107.6
## - Age                 1     1060 371794 1107.7
## - LastNightSleep      1     1064 371798 1107.7
## - Gamer               1     4366 375100 1108.8
## - PrimarilyHand       2    10827 381561 1109.0
## <none>                370734 1109.3
## - Sports_Frequency    1     5919 376653 1109.3
## - Distraction         1     9495 380229 1110.5
## - Shooting_Levels     2    16442 387176 1110.9
## - Game_Frequency      1    10647 381381 1110.9
## - Device              3    23274 394008 1111.1
## + logAvgHoursExercise 1      18 370717 1111.3

```

```

## - WiFi 1 13743 384477 1112.0
## + System 2 3475 367259 1112.1
## - Fatigue_3 2 21828 392562 1112.7
## - UsePrimaryHand 1 16130 386865 1112.8
## - Stress_3 2 26889 397623 1114.3
## + Cautious 4 7260 363474 1114.7
## - VisualAcuity 4 40924 411658 1114.8
## - AvgSleepTime 1 23825 394559 1115.3
## - CaffeinIntake 1 24115 394849 1115.4
## - Class 3 37930 408664 1115.8
## - RefreshRate 2 34559 405293 1116.8
## - InputDevice 4 105750 476484 1133.7
##
## Step: AIC=1107.31
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + Distraction +
## Temperature + Gamer + CaffeinIntake + VisualAcuity + PrimarilyHand +
## UsePrimaryHand + InputDevice + RefreshRate + Device + WiFi +
## Fatigue_3 + Stress_3 + Noise_Level + Game_Frequency + Sports_Frequency +
## Shooting_Levels
##
## Df Sum of Sq RSS AIC
## - Temperature 3 11970 382791 1105.4
## - Noise_Level 2 6813 377635 1105.7
## - LastNightSleep 1 1066 371888 1105.7
## - Age 1 1105 371926 1105.7
## - Gamer 1 4474 375295 1106.9
## <none> 370821 1107.3
## - Sports_Frequency 1 6209 377030 1107.5
## - PrimarilyHand 2 12830 383651 1107.7
## - Distraction 1 9417 380238 1108.5
## - Game_Frequency 1 10560 381381 1108.9
## - Shooting_Levels 2 16837 387659 1109.0
## - Device 3 23213 394034 1109.2
## + HoursAweak 1 87 370734 1109.3
## + logAvgHoursExercise 1 14 370807 1109.3
## - WiFi 1 13749 384571 1110.0
## + System 2 2952 367870 1110.3
## - Fatigue_3 2 23362 394183 1111.2
## - UsePrimaryHand 1 17655 388477 1111.3
## - Stress_3 2 27142 397963 1112.4
## + Cautious 4 7320 363501 1112.7
## - VisualAcuity 4 41189 412010 1112.9
## - AvgSleepTime 1 23826 394648 1113.3
## - CaffeinIntake 1 24468 395289 1113.6
## - Class 3 38459 409280 1114.0
## - RefreshRate 2 34613 405434 1114.8
## - InputDevice 4 107672 478493 1132.2
##
## Step: AIC=1105.41
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + Distraction +
## Gamer + CaffeinIntake + VisualAcuity + PrimarilyHand + UsePrimaryHand +
## InputDevice + RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 +
## Noise_Level + Game_Frequency + Sports_Frequency + Shooting_Levels
##

```

```

##              Df Sum of Sq    RSS    AIC
## - Noise_Level      2      4281 387073 1102.8
## - LastNightSleep    1       284 383075 1103.5
## - Age               1      1279 384070 1103.8
## - Gamer             1      1827 384618 1104.0
## - Sports_Frequency  1      5120 387912 1105.1
## <none>              382791 1105.4
## - Shooting_Levels   2     12603 395394 1105.6
## - Distraction        1      8186 390978 1106.1
## - Game_Frequency    1      8210 391001 1106.2
## - PrimarilyHand      2     16490 399282 1106.8
## - Device            3     23846 406638 1107.2
## + Temperature       3     11970 370821 1107.3
## + HoursAweak         1        97 382695 1107.4
## + logAvgHoursExercise 1        40 382751 1107.4
## + System             2      3673 379119 1108.2
## - WiFi              1     15089 397880 1108.4
## - Stress_3          2     23613 406404 1109.1
## - VisualAcuity       4     36566 419357 1109.2
## - Fatigue_3          2     28179 410971 1110.6
## - UsePrimaryHand     1     21981 404773 1110.6
## + Cautious           4      7923 374869 1110.7
## - AvgSleepTime       1     23256 406048 1111.0
## - Class              3     36590 419382 1111.2
## - CaffeinIntake      1     25129 407921 1111.6
## - RefreshRate        2     36072 418863 1113.0
## - InputDevice        4     100467 483258 1127.5
##
## Step:  AIC=1102.85
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + Distraction +
##           Gamer + CaffeinIntake + VisualAcuity + PrimarilyHand + UsePrimaryHand +
##           InputDevice + RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 +
##           Game_Frequency + Sports_Frequency + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - LastNightSleep    1       383 387456 1101.0
## - Gamer             1      1318 388391 1101.3
## - Age               1      1614 388687 1101.4
## - Shooting_Levels   2     11193 398266 1102.5
## <none>              387073 1102.8
## - Distraction        1      6132 393205 1102.9
## - Sports_Frequency   1      6408 393481 1103.0
## - Game_Frequency     1      8205 395278 1103.5
## - PrimarilyHand      2     16349 403422 1104.2
## + HoursAweak         1        74 386999 1104.8
## + logAvgHoursExercise 1         1 387071 1104.8
## + Noise_Level        2      4281 382791 1105.4
## - WiFi              1     14680 401753 1105.7
## + Temperature       3      9438 377635 1105.7
## + System             2      2576 384497 1106.0
## - Stress_3          2     23485 410557 1106.4
## - Device            3     31250 418322 1106.9
## - VisualAcuity       4     38848 425921 1107.2
## + Cautious           4      9580 377493 1107.6

```

```

## - Fatigue_3          2      27266 414338 1107.6
## - Class              3      34189 421262 1107.8
## - AvgSleepTime       1      23007 410080 1108.3
## - CaffeinIntake       1      23850 410922 1108.6
## - UsePrimaryHand      1      24556 411629 1108.8
## - RefreshRate         2      35695 422768 1110.2
## - InputDevice         4      98624 485696 1124.1
##
## Step: AIC=1100.97
## avg_time ~ Class + Age + AvgSleepTime + Distraction + Gamer +
##   CaffeinIntake + VisualAcuity + PrimarilyHand + UsePrimaryHand +
##   InputDevice + RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 +
##   Game_Frequency + Sports_Frequency + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - Gamer          1      1334 388790 1099.4
## - Age            1      1828 389284 1099.6
## - Shooting_Levels 2     11178 398633 1100.6
## - Distraction     1      5951 393406 1100.9
## <none>              387456 1101.0
## - Sports_Frequency 1      6316 393772 1101.1
## - Game_Frequency  1      8139 395594 1101.7
## - PrimarilyHand    2     16257 403712 1102.3
## + LastNightSleep   1       383 387073 1102.8
## + HoursAweak        1        76 387380 1103.0
## + logAvgHoursExercise 1         17 387438 1103.0
## + Noise_Level       2      4380 383075 1103.5
## - WiFi              1     14370 401826 1103.7
## + Temperature       3      8701 378754 1104.0
## + System             2      2691 384764 1104.1
## - Stress_3          2     23111 410566 1104.5
## - Device            3     31664 419119 1105.1
## - VisualAcuity       4     38508 425964 1105.2
## + Cautious          4      9643 377812 1105.7
## - Class             3     34943 422398 1106.1
## - CaffeinIntake      1     23961 411416 1106.7
## - Fatigue_3         2     30543 417998 1106.8
## - UsePrimaryHand     1     24445 411900 1106.9
## - RefreshRate        2     35509 422965 1108.3
## - AvgSleepTime       1     29291 416746 1108.4
## - InputDevice        4     100671 488127 1122.8
##
## Step: AIC=1099.42
## avg_time ~ Class + Age + AvgSleepTime + Distraction + CaffeinIntake +
##   VisualAcuity + PrimarilyHand + UsePrimaryHand + InputDevice +
##   RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Game_Frequency +
##   Sports_Frequency + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - Age            1      2399 391188 1098.2
## - Shooting_Levels 2     10171 398960 1098.8
## - Distraction     1      5525 394315 1099.2
## - Sports_Frequency 1      5816 394606 1099.3
## <none>              388790 1099.4

```



```

## - Game_Frequency      1      6920 395710 1099.7
## - PrimarilyHand       2     15844 404634 1100.6
## + Gamer               1      1334 387456 1101.0
## + LastNightSleep      1       399 388391 1101.3
## + HoursAweak          1       123 388667 1101.4
## + logAvgHoursExercise  1        55 388734 1101.4
## - WiFi                1     13216 402006 1101.7
## + Noise_Level         2      3866 384924 1102.1
## + System              2      2804 385985 1102.5
## - Stress_3            2     22081 410871 1102.5
## + Temperature         3      6969 381821 1103.1
## - VisualAcuity        4     37179 425969 1103.2
## - Device              3     31474 420264 1103.5
## + Cautious            4      9881 378909 1104.1
## - Class               3     33671 422461 1104.1
## - CaffeinIntake       1     22733 411522 1104.8
## - Fatigue_3           2     29498 418288 1104.8
## - UsePrimaryHand      1     23127 411916 1104.9
## - AvgSleepTime        1     28824 417614 1106.6
## - RefreshRate         2     35662 424452 1106.7
## - InputDevice         4    100536 489325 1121.1
##
## Step: AIC=1098.21
## avg_time ~ Class + AvgSleepTime + Distraction + CaffeinIntake +
##     VisualAcuity + PrimarilyHand + UsePrimaryHand + InputDevice +
##     RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Game_Frequency +
##     Sports_Frequency + Shooting_Levels
##
##              Df Sum of Sq   RSS   AIC
## - Shooting_Levels      2      9379 400568 1097.3
## - Distraction           1      5870 397058 1098.1
## <none>                  391188 1098.2
## - Sports_Frequency      1      7300 398489 1098.6
## - Game_Frequency        1      7549 398738 1098.7
## + Age                   1      2399 388790 1099.4
## + Gamer                 1      1905 389284 1099.6
## - PrimarilyHand         2     17520 408708 1099.9
## + LastNightSleep        1       661 390527 1100.0
## - WiFi                  1     11941 403130 1100.1
## + logAvgHoursExercise   1       275 390913 1100.1
## + HoursAweak            1       211 390977 1100.1
## + Noise_Level           2      4203 386986 1100.8
## + System                2      3594 387594 1101.0
## - Stress_3              2     22189 413377 1101.3
## - VisualAcuity          4     35439 426628 1101.4
## + Temperature           3      6681 384507 1102.0
## - Device                3     32199 423387 1102.4
## + Cautious              4     10431 380757 1102.7
## - Fatigue_3             2     30846 422034 1104.0
## - UsePrimaryHand        1     25826 417014 1104.5
## - CaffeinIntake         1     25935 417123 1104.5
## - AvgSleepTime          1     26988 418176 1104.8
## - RefreshRate           2     35935 427124 1105.5
## - Class                 3     80062 471250 1116.2

```

```

## - InputDevice          4      99795 490984 1119.5
##
## Step: AIC=1097.27
## avg_time ~ Class + AvgSleepTime + Distraction + CaffeinIntake +
##      VisualAcuity + PrimarilyHand + UsePrimaryHand + InputDevice +
##      RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Game_Frequency +
##      Sports_Frequency
##
##              Df Sum of Sq    RSS    AIC
## - Distraction      1      5333 405900 1097.0
## <none>                      400568 1097.3
## - WiFi              1      8805 409372 1098.1
## - PrimarilyHand     2     15495 416062 1098.2
## + Shooting_Levels   2      9379 391188 1098.2
## - Sports_Frequency  1     10001 410569 1098.5
## + Age               1      1607 398960 1098.8
## + logAvgHoursExercise 1      1285 399283 1098.8
## + Gamer             1       605 399962 1099.1
## + LastNightSleep    1       592 399976 1099.1
## + HoursAweak        1       476 400092 1099.1
## - Game_Frequency    1     12442 413010 1099.2
## + System            2       4113 396455 1099.9
## + Noise_Level       2       2983 397584 1100.3
## - Stress_3          2     24562 425129 1100.9
## + Cautious          4     11847 388721 1101.4
## + Temperature       3       4947 395620 1101.7
## - VisualAcuity      4     40553 441121 1101.7
## - UsePrimaryHand    1     20795 421363 1101.8
## - Device            3     37731 438298 1102.9
## - CaffeinIntake     1     25512 426080 1103.2
## - AvgSleepTime      1     25819 426386 1103.3
## - Fatigue_3         2     39403 439970 1105.4
## - RefreshRate       2     40891 441458 1105.8
## - Class             3     94721 495289 1118.7
## - InputDevice       4     111509 512077 1121.0
##
## Step: AIC=1096.97
## avg_time ~ Class + AvgSleepTime + CaffeinIntake + VisualAcuity +
##      PrimarilyHand + UsePrimaryHand + InputDevice + RefreshRate +
##      Device + WiFi + Fatigue_3 + Stress_3 + Game_Frequency + Sports_Frequency
##
##              Df Sum of Sq    RSS    AIC
## <none>                      405900 1097.0
## - PrimarilyHand     2     13237 419137 1097.1
## + Distraction       1      5333 400568 1097.3
## - WiFi              1      8406 414306 1097.6
## - Sports_Frequency  1      8986 414886 1097.8
## + Shooting_Levels   2     8842 397058 1098.1
## + Age               1      1943 403957 1098.3
## + logAvgHoursExercise 1      1905 403995 1098.4
## - Game_Frequency    1     11911 417811 1098.7
## + LastNightSleep    1       428 405473 1098.8
## + HoursAweak        1       399 405501 1098.8
## + Gamer             1       398 405503 1098.8

```

```
## + System                2      3809 402091 1099.8
## + Noise_Level            2      1132 404768 1100.6
## + Cautious               4     12107 393793 1101.1
## + Temperature            3      4990 400910 1101.4
## - UsePrimaryHand         1     20798 426698 1101.4
## - Stress_3               2     28495 434395 1101.7
## - VisualAcuity           4     43516 449416 1102.1
## - AvgSleepTime           1     24693 430593 1102.6
## - Device                 3     38320 444220 1102.6
## - CaffeinIntake          1     33563 439463 1105.2
## - Fatigue_3              2     41583 447483 1105.5
## - RefreshRate            2     42714 448614 1105.9
## - Class                  3     93752 499652 1117.8
## - InputDevice            4    115093 520993 1121.2
```

```
##
```

```
## Call:
```

```
## lm(formula = avg_time ~ Class + AvgSleepTime + CaffeinIntake +
##      VisualAcuity + PrimarilyHand + UsePrimaryHand + InputDevice +
##      RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Game_Frequency +
##      Sports_Frequency, data = Survey_cleaned)
```

```
##
```

```
## Coefficients:
```

```
##                      (Intercept)
##                      438.055
##                      ClassJunior
##                      38.454
##                      ClassSenior
##                      15.073
##                      ClassGraduate student
##                      129.289
##                      AvgSleepTime
##                      15.025
##                      CaffeinIntakeYes
##                      43.014
##                      VisualAcuityExcellent
##                      -69.927
##                      VisualAcuityGood
##                      -83.490
##                      VisualAcuityPoor
##                      -73.772
##                      VisualAcuityVery Poor
##                      -116.392
##                      PrimarilyHandLeft hand
##                      -51.238
## PrimarilyHandAmbidextrous (both hands equally)
##                      31.844
##                      UsePrimaryHandNo
##                      74.840
##                      InputDeviceMouse
##                      -51.908
##                      InputDeviceKeyboard
##                      53.504
##                      InputDeviceTrackpad
```

```

##                                25.861
##          InputDeviceTouch screen
##                                -20.490
##          RefreshRateMid-Range
##                                27.233
##          RefreshRateStandard
##                                57.230
##          DeviceiPad(Tablet
##                                -24.929
##          DeviceiPhone(Smartphone
##                                142.767
##          DeviceLaptop
##                                36.504
##          WiFiStable
##                                -71.611
##          Fatigue_3Moderately Fatigue
##                                63.388
##          Fatigue_3Low Fatigue
##                                56.646
##          Stress_3Moderately Stress
##                                7.428
##          Stress_3Low Stress
##                                -32.981
##          Game_FrequencyLow Frequency
##                                21.879
##          Sports_FrequencyLow Frequency
##                                20.661

## Start:  AIC=1257.15
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Temperature + Gamer + CaffeinIntake + VisualAcuity +
##   PrimarilyHand + UsePrimaryHand + Cautious + InputDevice +
##   RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Noise_Level +
##   Game_Frequency + Sports_Frequency + Shooting_Levels + System +
##   logAvgHoursExercise
##
##           Df Sum of Sq   RSS   AIC
## - Cautious      4      6186 367259 1239.9
## - Temperature    3     10845 371918 1246.4
## - System          2      1980 363052 1248.1
## - Noise_Level     2      5074 366147 1249.2
## - VisualAcuity    4      37145 398218 1250.3
## - Device          3     22755 383828 1250.5
## - PrimarilyHand   2     11526 372599 1251.5
## - Shooting_Levels  2     11896 372969 1251.6
## - logAvgHoursExercise 1       273 361346 1252.4
## - HoursAweak      1       301 361374 1252.4
## - LastNightSleep   1       493 361565 1252.5
## - Age              1       626 361699 1252.5
## - Gamer            1      3273 364346 1253.5
## - Sports_Frequency  1      4170 365243 1253.8
## - Fatigue_3        2     19740 380813 1254.3
## - Distraction      1      8935 370008 1255.4
## - Stress_3         2     23328 384401 1255.5

```

```

## - Game_Frequency      1      9347 370420 1255.6
## - Class                3      40144 401217 1256.2
## - UsePrimaryHand       1      11674 372746 1256.4
## - WiFi                 1      12234 373307 1256.6
## <none>                  361073 1257.2
## - AvgSleepTime         1      15842 376915 1257.8
## - RefreshRate          2      34921 395994 1259.3
## - CaffeinIntake        1      22272 383345 1260.0
## - InputDevice          4      73291 434364 1261.5
##
## Step: AIC=1239.9
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Temperature + Gamer + CaffeinIntake + VisualAcuity +
##   PrimarilyHand + UsePrimaryHand + InputDevice + RefreshRate +
##   Device + WiFi + Fatigue_3 + Stress_3 + Noise_Level + Game_Frequency +
##   Sports_Frequency + Shooting_Levels + System + logAvgHoursExercise
##
##              Df Sum of Sq    RSS    AIC
## - Temperature      3      11208 378467 1229.2
## - System            2       3458 370717 1231.4
## - Device            3      22527 389786 1233.0
## - Noise_Level       2       8204 375463 1233.0
## - VisualAcuity      4      39758 407017 1233.7
## - PrimarilyHand     2      10345 377604 1233.8
## - logAvgHoursExercise 1         0 367259 1235.0
## - Shooting_Levels   2      14422 381681 1235.2
## - Age               1       450 367708 1235.2
## - LastNightSleep    1       519 367778 1235.2
## - HoursAweak        1       611 367869 1235.2
## - Gamer             1       3617 370875 1236.3
## - Sports_Frequency  1       4109 371368 1236.5
## - Fatigue_3         2      23869 391128 1238.3
## - Stress_3          2      23955 391213 1238.3
## - Distraction       1      10066 377324 1238.5
## - Class             3      40028 407286 1238.7
## - Game_Frequency    1      10895 378153 1238.8
## - WiFi              1      12477 379736 1239.3
## <none>              367259 1239.9
## - UsePrimaryHand    1      15475 382733 1240.4
## - RefreshRate       2      36741 404000 1242.5
## - CaffeinIntake     1      22391 389650 1242.7
## - AvgSleepTime      1      26198 393457 1243.9
## - InputDevice       4      74330 441589 1244.2
## + Cautious          4       6186 361073 1257.2
##
## Step: AIC=1229.2
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Gamer + CaffeinIntake + VisualAcuity + PrimarilyHand +
##   UsePrimaryHand + InputDevice + RefreshRate + Device + WiFi +
##   Fatigue_3 + Stress_3 + Noise_Level + Game_Frequency + Sports_Frequency +
##   Shooting_Levels + System + logAvgHoursExercise
##
##              Df Sum of Sq    RSS    AIC
## - System            2       4184 382651 1220.9

```

```

## - VisualAcuity      4      35557 414024 1221.3
## - Noise_Level      2        5619 384085 1221.4
## - Device           3      23281 401748 1222.3
## - Shooting_Levels  2      10787 389254 1223.1
## - PrimarilyHand    2      13084 391551 1223.9
## - logAvgHoursExercise 1         4 378471 1224.3
## - LastNightSleep   1        78 378545 1224.4
## - Age              1        538 379005 1224.5
## - HoursAweak       1        650 379117 1224.6
## - Gamer            1       1425 379892 1224.8
## - Sports_Frequency 1       3391 381858 1225.5
## - Stress_3         2      20992 399459 1226.4
## - Class            3      38229 416696 1227.0
## - Game_Frequency   1       8495 386962 1227.2
## - Distraction       1       8793 387260 1227.3
## - Fatigue_3        2      27884 406351 1228.7
## - WiFi             1      13749 392215 1228.9
## <none>              378467 1229.2
## - UsePrimaryHand   1      19388 397855 1230.8
## - InputDevice      4      68310 446777 1231.2
## - CaffeinIntake     1      22626 401093 1231.8
## - RefreshRate       2      38413 416880 1232.0
## - AvgSleepTime     1      25529 403996 1232.8
## + Temperature      3      11208 367259 1239.9
## + Cautious         4       6549 371918 1246.4
##
## Step:  AIC=1220.9
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Gamer + CaffeinIntake + VisualAcuity + PrimarilyHand +
##   UsePrimaryHand + InputDevice + RefreshRate + Device + WiFi +
##   Fatigue_3 + Stress_3 + Noise_Level + Game_Frequency + Sports_Frequency +
##   Shooting_Levels + logAvgHoursExercise
##
##              Df Sum of Sq  RSS    AIC
## - Noise_Level      2      4347 386998 1212.6
## - VisualAcuity      4     36345 418996 1213.2
## - Device           3     23904 406555 1214.1
## - Shooting_Levels  2     11863 394514 1215.1
## - PrimarilyHand    2     13695 396346 1215.7
## - logAvgHoursExercise 1        44 382695 1216.0
## - HoursAweak       1        101 382751 1216.1
## - LastNightSleep   1        309 382960 1216.1
## - Age              1       1275 383925 1216.5
## - Gamer            1       1799 384450 1216.6
## - Sports_Frequency  1       4666 387317 1217.6
## - Class            3     36049 418700 1217.9
## - Game_Frequency   1       8180 390831 1218.8
## - Distraction       1       8254 390905 1218.8
## - Stress_3         2     23453 406104 1218.8
## - Fatigue_3        2     25061 407712 1219.4
## <none>              382651 1220.9
## - WiFi             1     15046 397697 1221.0
## - RefreshRate       2     34523 417174 1222.3
## - UsePrimaryHand   1     20006 402657 1222.6

```

```

## - AvgSleepTime      1      23174 405825 1223.6
## - CaffeinIntake     1      24173 406824 1223.9
## + System            2         4184 378467 1229.2
## - InputDevice       4      98241 480891 1230.9
## + Temperature       3      11934 370717 1231.4
## + Cautious          4         7997 374654 1237.6
##
## Step: AIC=1212.63
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Gamer + CaffeinIntake + VisualAcuity + PrimarilyHand +
##   UsePrimaryHand + InputDevice + RefreshRate + Device + WiFi +
##   Fatigue_3 + Stress_3 + Game_Frequency + Sports_Frequency +
##   Shooting_Levels + logAvgHoursExercise
##
##              Df Sum of Sq   RSS   AIC
## - VisualAcuity      4      38779 425777 1205.5
## - Shooting_Levels    2      10264 397262 1206.3
## - PrimarilyHand      2      13663 400661 1207.4
## - logAvgHoursExercise 1           1 386999 1207.8
## - HoursAweak         1          73 387071 1207.8
## - LastNightSleep     1          367 387365 1207.9
## - Device             3      31210 418208 1208.1
## - Gamer              1          1257 388255 1208.2
## - Age               1          1519 388517 1208.3
## - Class             3      33787 420785 1208.8
## - Sports_Frequency   1          5428 392426 1209.6
## - Distraction        1          6008 393006 1209.8
## - Stress_3          2      23074 410072 1210.4
## - Game_Frequency     1          8251 395250 1210.5
## - Fatigue_3         2      23851 410849 1210.6
## - WiFi              1      14667 401666 1212.6
## <none>                386998 1212.6
## - RefreshRate       2      34545 421543 1213.9
## - UsePrimaryHand     1      22667 409665 1215.1
## - AvgSleepTime       1      23015 410013 1215.2
## - CaffeinIntake      1      23215 410213 1215.3
## + Noise_Level       2          4347 382651 1220.9
## + System            2          2913 384085 1221.4
## - InputDevice       4      96745 483743 1222.0
## + Temperature       3          9469 377529 1224.0
## + Cautious          4          9588 377410 1228.8
##
## Step: AIC=1205.51
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Gamer + CaffeinIntake + PrimarilyHand + UsePrimaryHand +
##   InputDevice + RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 +
##   Game_Frequency + Sports_Frequency + Shooting_Levels + logAvgHoursExercise
##
##              Df Sum of Sq   RSS   AIC
## - Class             3      30472 456249 1199.8
## - Shooting_Levels    2      13920 439697 1199.9
## - Stress_3          2      14882 440659 1200.2
## - Gamer             1           5 425782 1200.7
## - logAvgHoursExercise 1          20 425797 1200.7

```

```

## - LastNightSleep      1      55 425832 1200.7
## - HoursAweak          1     126 425903 1200.7
## - Age                 1     547 426324 1200.8
## - PrimarilyHand       2    16925 442702 1200.8
## - Device              3    36091 461868 1201.4
## - Sports_Frequency    1     4645 430422 1202.0
## - Game_Frequency      1     6796 432573 1202.7
## - WiFi                1     8600 434376 1203.2
## - Distraction         1     8767 434544 1203.3
## - Fatigue_3           2    29208 454985 1204.3
## - CaffeinIntake       1    14298 440074 1204.9
## <none>                1     425777 1205.5
## - RefreshRate         2    36358 462134 1206.4
## - AvgSleepTime        1    24160 449937 1207.8
## - UsePrimaryHand      1    26149 451926 1208.3
## + VisualAcuity        4    38779 386998 1212.6
## + Noise_Level         2     6781 418996 1213.2
## + System              2     3121 422656 1214.3
## - InputDevice         4   111805 537582 1216.2
## + Temperature         3     5643 420134 1218.4
## + Cautious            4    12690 413087 1221.0
##
## Step:  AIC=1199.85
## avg_time ~ Age + AvgSleepTime + LastNightSleep + HoursAweak +
##      Distraction + Gamer + CaffeinIntake + PrimarilyHand + UsePrimaryHand +
##      InputDevice + RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 +
##      Game_Frequency + Sports_Frequency + Shooting_Levels + logAvgHoursExercise
##
##              Df Sum of Sq  RSS    AIC
## - PrimarilyHand      2    15200 471449 1194.4
## - Device              3    34447 490696 1194.7
## - logAvgHoursExercise 1     179 456428 1195.0
## - LastNightSleep     1     254 456503 1195.1
## - HoursAweak         1     318 456567 1195.1
## - Gamer              1     525 456774 1195.1
## - Sports_Frequency    1     3518 459767 1196.0
## - Shooting_Levels     2    21649 477898 1196.1
## - Game_Frequency      1     4527 460777 1196.3
## - Fatigue_3          2    23691 479940 1196.7
## - CaffeinIntake       1     7922 464172 1197.2
## - Distraction         1     9878 466127 1197.8
## - WiFi                1    10276 466525 1197.9
## - Stress_3            2    31047 487296 1198.6
## - RefreshRate         2    32563 488812 1199.0
## <none>                1     456249 1199.8
## - UsePrimaryHand      1    26461 482710 1202.3
## - AvgSleepTime        1    30040 486289 1203.2
## + Class               3    30472 425777 1205.5
## - InputDevice         4   102228 558477 1206.5
## - Age                 1    42831 499080 1206.6
## + Noise_Level         2     4071 452178 1208.4
## + VisualAcuity        4    35464 420785 1208.8
## + System              2     1296 454953 1209.2
## + Temperature         3     5658 450591 1212.8

```



```

## + Cautious          4      11104 445145 1216.1
##
## Step: AIC=1194.36
## avg_time ~ Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Gamer + CaffeinIntake + UsePrimaryHand + InputDevice +
##   RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Game_Frequency +
##   Sports_Frequency + Shooting_Levels + logAvgHoursExercise
##
##           Df Sum of Sq    RSS    AIC
## - Device      3      24984 496434 1186.4
## - Shooting_Levels  2      17731 489180 1189.4
## - logAvgHoursExercise  1         35 471484 1189.5
## - LastNightSleep    1        138 471587 1189.5
## - Sports_Frequency   1        909 472359 1189.8
## - Gamer              1        921 472370 1189.8
## - HoursAweak         1       3828 475278 1190.5
## - Game_Frequency     1       4999 476449 1190.9
## - CaffeinIntake      1       6754 478203 1191.3
## - Distraction        1       6974 478423 1191.4
## - WiFi              1       9910 481360 1192.2
## - RefreshRate        2      30567 502017 1192.7
## - Fatigue_3          2      34343 505793 1193.7
## - UsePrimaryHand     1      17381 488830 1194.2
## <none>                        471449 1194.4
## - Stress_3          2      41305 512754 1195.5
## - AvgSleepTime      1      28267 499716 1197.0
## + PrimarilyHand     2      15200 456249 1199.8
## - Age               1      43111 514560 1200.8
## + Class              3      28747 442702 1200.8
## - InputDevice        4     110627 582077 1202.1
## + Noise_Level        2        5782 465667 1202.5
## + VisualAcuity       4      38136 433314 1202.9
## + System             2        2180 469269 1203.5
## + Temperature        3        7900 463549 1206.8
## + Cautious           4       13026 458423 1210.2
##
## Step: AIC=1186.44
## avg_time ~ Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Gamer + CaffeinIntake + UsePrimaryHand + InputDevice +
##   RefreshRate + WiFi + Fatigue_3 + Stress_3 + Game_Frequency +
##   Sports_Frequency + Shooting_Levels + logAvgHoursExercise
##
##           Df Sum of Sq    RSS    AIC
## - logAvgHoursExercise  1         17 496451 1181.6
## - LastNightSleep      1         482 496916 1181.7
## - Sports_Frequency     1         771 497205 1181.8
## - Gamer                1        1229 497663 1181.9
## - HoursAweak           1        2382 498815 1182.2
## - Game_Frequency       1        2441 498874 1182.2
## - Shooting_Levels      2       25184 521618 1183.1
## - CaffeinIntake        1        8974 505407 1183.9
## - Distraction          1       10184 506618 1184.2
## - Fatigue_3            2       30325 526759 1184.4
## - WiFi                 1       12878 509311 1184.9

```

```

## - UsePrimaryHand      1      15112 511545 1185.5
## <none>                  496434 1186.4
## - RefreshRate         2      43484 539917 1187.5
## - Stress_3            2      47521 543954 1188.5
## - AvgSleepTime        1      31235 527669 1189.5
## - Age                 1      38244 534678 1191.2
## + Class               3      28012 468421 1193.5
## + Device              3      24984 471449 1194.4
## + PrimarilyHand       2        5737 490696 1194.7
## + Noise_Level         2        5195 491238 1194.8
## + VisualAcuity        4      39298 457135 1195.2
## + System              2        1585 494848 1195.8
## + Temperature        3        6437 489997 1199.3
## - InputDevice         4     141996 638430 1199.5
## + Cautious            4       16082 480351 1201.6
##
## Step:  AIC=1181.59
## avg_time ~ Age + AvgSleepTime + LastNightSleep + HoursAweak +
##      Distraction + Gamer + CaffeinIntake + UsePrimaryHand + InputDevice +
##      RefreshRate + WiFi + Fatigue_3 + Stress_3 + Game_Frequency +
##      Sports_Frequency + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - LastNightSleep      1         540 496991 1176.9
## - Sports_Frequency     1         958 497408 1177.0
## - Gamer                1        1217 497668 1177.0
## - HoursAweak          1        2420 498871 1177.3
## - Game_Frequency       1        2427 498877 1177.3
## - Shooting_Levels      2       26592 523043 1178.6
## - CaffeinIntake        1        9107 505558 1179.1
## - Distraction          1       10562 507013 1179.4
## - Fatigue_3            2       31573 528023 1179.8
## - WiFi                 1       12865 509316 1180.0
## - UsePrimaryHand       1       15160 511611 1180.6
## <none>                  496451 1181.6
## - RefreshRate         2       44380 540831 1182.9
## - Stress_3            2       47944 544395 1183.8
## - AvgSleepTime        1       31500 527951 1184.7
## + logAvgHoursExercise  1          17 496434 1186.4
## - Age                 1       40034 536485 1186.7
## + Class               3       27886 468565 1188.7
## + Device              3       24967 471484 1189.5
## + PrimarilyHand       2        5594 490856 1189.8
## + Noise_Level         2        5212 491238 1189.9
## + VisualAcuity        4       39221 457230 1190.4
## + System              2        1550 494900 1190.9
## + Temperature        3        6451 489999 1194.5
## - InputDevice         4     142428 638879 1194.7
## + Cautious            4       16031 480419 1196.8
##
## Step:  AIC=1176.87
## avg_time ~ Age + AvgSleepTime + HoursAweak + Distraction + Gamer +
##      CaffeinIntake + UsePrimaryHand + InputDevice + RefreshRate +
##      WiFi + Fatigue_3 + Stress_3 + Game_Frequency + Sports_Frequency +

```

```

##      Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - Sports_Frequency  1      917 497908 1172.2
## - Gamer              1     1223 498214 1172.3
## - Game_Frequency    1     2348 499339 1172.6
## - HoursAweak        1     2452 499443 1172.6
## - Shooting_Levels   2    26600 523591 1173.9
## - CaffeinIntake     1     9098 506089 1174.3
## - Distraction       1    10276 507267 1174.7
## - WiFi              1    12476 509467 1175.2
## - Fatigue_3         2    34112 531103 1175.7
## - UsePrimaryHand    1    15027 512017 1175.8
## <none>                496991 1176.9
## - RefreshRate       2    44203 541193 1178.1
## - Stress_3          2    47405 544396 1178.9
## + LastNightSleep    1      540 496451 1181.6
## + logAvgHoursExercise 1       75 496916 1181.7
## - AvgSleepTime      1    39943 536934 1182.0
## - Age               1    41602 538593 1182.4
## + Class              3    28221 468770 1183.9
## + Device             3    25335 471656 1184.7
## + PrimarilyHand     2     5551 491440 1185.1
## + Noise_Level       2     5306 491685 1185.2
## + VisualAcuity      4    38058 458933 1186.0
## + System            2     1434 495557 1186.2
## + Temperature       3     6038 490953 1189.9
## - InputDevice       4    146159 643150 1190.7
## + Cautious          4     15922 481069 1192.1
##
## Step:  AIC=1172.24
## avg_time ~ Age + AvgSleepTime + HoursAweak + Distraction + Gamer +
##      CaffeinIntake + UsePrimaryHand + InputDevice + RefreshRate +
##      WiFi + Fatigue_3 + Stress_3 + Game_Frequency + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - Gamer              1     1386 499295 1167.7
## - HoursAweak        1     2721 500630 1168.1
## - Game_Frequency    1     3143 501051 1168.2
## - CaffeinIntake     1     8335 506243 1169.5
## - Shooting_Levels   2    28238 526146 1169.6
## - Distraction       1    10235 508143 1170.0
## - WiFi              1    12099 510007 1170.5
## - Fatigue_3         2    33464 531372 1170.9
## - UsePrimaryHand    1    15333 513241 1171.3
## <none>                497908 1172.2
## - RefreshRate       2    43934 541842 1173.4
## - Stress_3          2    49014 546922 1174.6
## + Sports_Frequency  1      917 496991 1176.9
## + LastNightSleep    1      500 497408 1177.0
## + logAvgHoursExercise 1      325 497583 1177.0
## - AvgSleepTime      1    39834 537743 1177.3
## - Age               1    42854 540762 1178.0
## + Class              3    27684 470225 1179.4

```

```

## + Device          3      25150 472758 1180.1
## + Noise_Level     2       5635 492273 1180.5
## + PrimarilyHand   2       4831 493077 1180.7
## + System          2       1516 496392 1181.6
## + VisualAcuity    4      37485 460423 1181.6
## + Temperature     3       5636 492273 1185.4
## - InputDevice     4     145248 643156 1185.8
## + Cautious        4       16828 481080 1187.2
##
## Step: AIC=1167.74
## avg_time ~ Age + AvgSleepTime + HoursAweak + Distraction + CaffeinIntake +
##      UsePrimaryHand + InputDevice + RefreshRate + WiFi + Fatigue_3 +
##      Stress_3 + Game_Frequency + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - HoursAweak      1      2588 501882 1163.5
## - Game_Frequency   1      5180 504475 1164.2
## - CaffeinIntake    1     10350 509645 1165.5
## - Distraction      1     10866 510161 1165.7
## - Shooting_Levels  2     33060 532355 1166.3
## - Fatigue_3        2     33861 533156 1166.5
## - WiFi             1     14221 513515 1166.5
## - UsePrimaryHand   1     18935 518229 1167.7
## <none>              499295 1167.7
## - RefreshRate      2     43903 543198 1168.9
## - Stress_3         2     51385 550680 1170.7
## + Gamer            1      1386 497908 1172.2
## + Sports_Frequency 1      1081 498214 1172.3
## + LastNightSleep   1       503 498792 1172.5
## + logAvgHoursExercise 1       290 499005 1172.5
## - AvgSleepTime     1     39965 539260 1172.8
## - Age              1     43453 542747 1173.7
## + Class            3     28805 470489 1174.7
## + Device           3     25513 473781 1175.6
## + Noise_Level      2      6074 493221 1175.9
## + PrimarilyHand    2      5074 494220 1176.1
## + VisualAcuity     4     38863 460432 1176.7
## + System           2      1724 497571 1177.0
## + Temperature      3      6500 492795 1180.6
## + Cautious         4     15940 483355 1183.0
## - InputDevice      4     155703 654997 1183.3
##
## Step: AIC=1163.55
## avg_time ~ Age + AvgSleepTime + Distraction + CaffeinIntake +
##      UsePrimaryHand + InputDevice + RefreshRate + WiFi + Fatigue_3 +
##      Stress_3 + Game_Frequency + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - Game_Frequency   1      4791 506674 1159.9
## - Distraction      1      9976 511858 1161.2
## - CaffeinIntake    1     10020 511902 1161.2
## - Fatigue_3        2     31583 533465 1161.7
## - WiFi             1     13625 515507 1162.2
## - Shooting_Levels  2     33587 535469 1162.2

```

```

## <none>                                501882 1163.5
## - UsePrimaryHand      1      20303 522185 1163.8
## - RefreshRate         2      41954 543837 1164.2
## - Stress_3            2      49054 550936 1165.9
## + HoursAweak          1       2588 499295 1167.7
## + Sports_Frequency    1       1354 500528 1168.1
## + Gamer                1       1253 500630 1168.1
## - AvgSleepTime        1      38386 540268 1168.2
## + LastNightSleep      1        528 501354 1168.3
## + logAvgHoursExercise 1        469 501413 1168.3
## - Age                 1      43493 545375 1169.4
## + Class               3      29478 472404 1170.3
## + PrimarilyHand       2       6881 495002 1171.5
## + Noise_Level         2       5836 496046 1171.8
## + Device              3      23758 478124 1171.9
## + VisualAcuity        4      38846 463036 1172.6
## + System              2       2226 499657 1172.7
## + Temperature         3       6485 495397 1176.5
## + Cautious            4      16785 485098 1178.6
## - InputDevice         4     156592 658474 1179.1
##
## Step: AIC=1159.92
## avg_time ~ Age + AvgSleepTime + Distraction + CaffeinIntake +
##      UsePrimaryHand + InputDevice + RefreshRate + WiFi + Fatigue_3 +
##      Stress_3 + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - Distraction      1      9399 516073 1157.4
## - Fatigue_3        2     30803 537476 1157.8
## - CaffeinIntake     1     11427 518101 1157.9
## - WiFi              1     17303 523976 1159.4
## <none>                                506674 1159.9
## - Shooting_Levels  2     40965 547638 1160.2
## - UsePrimaryHand   1     21635 528308 1160.5
## - RefreshRate      2     42254 548928 1160.5
## - Stress_3         2     52770 559444 1163.0
## + Game_Frequency   1       4791 501882 1163.5
## + Gamer            1       3135 503539 1164.0
## + Sports_Frequency 1       2730 503944 1164.1
## - AvgSleepTime     1     36788 543462 1164.1
## + HoursAweak       1       2199 504475 1164.2
## + logAvgHoursExercise 1        480 506193 1164.7
## + LastNightSleep   1        393 506281 1164.7
## - Age              1     47627 554301 1166.7
## + Class            3     28130 478544 1167.1
## + PrimarilyHand    2       7216 499457 1167.8
## + Noise_Level      2       5861 500812 1168.1
## + System           2       3106 503568 1168.8
## + VisualAcuity     4     39132 467542 1169.0
## + Device           3     20460 486214 1169.2
## + Temperature      3       5374 501300 1173.1
## + Cautious         4     15584 491090 1175.3
## - InputDevice      4     161112 667786 1176.1
##

```

```

## Step: AIC=1157.43
## avg_time ~ Age + AvgSleepTime + CaffeinIntake + UsePrimaryHand +
##      InputDevice + RefreshRate + WiFi + Fatigue_3 + Stress_3 +
##      Shooting_Levels
##
##      Df Sum of Sq    RSS    AIC
## - Fatigue_3      2    32434 548507 1155.6
## - WiFi           1    16084 532157 1156.5
## - CaffeinIntake   1    18303 534376 1157.1
## <none>                        516073 1157.4
## - Shooting_Levels 2    40953 557026 1157.6
## - RefreshRate     2    44636 560709 1158.4
## - UsePrimaryHand  1    25136 541209 1158.7
## + Distraction     1     9399 506674 1159.9
## - AvgSleepTime    1    34735 550808 1161.0
## + Game_Frequency  1     4215 511858 1161.2
## + Gamer           1     3809 512264 1161.3
## + Sports_Frequency 1     2540 513533 1161.7
## + HoursAweak      1     1426 514647 1161.9
## + logAvgHoursExercise 1     1099 514974 1162.0
## + LastNightSleep  1       167 515906 1162.2
## - Stress_3        2    62638 578711 1162.5
## - Age             1    45851 561924 1163.5
## + Class           3    29900 486173 1164.3
## + VisualAcuity    4    43461 472612 1165.5
## + PrimarilyHand   2     5354 510719 1165.8
## + Device          3    23544 492529 1166.0
## + System          2     4423 511650 1166.0
## + Noise_Level     2     3899 512174 1166.2
## + Temperature     3     5094 510979 1170.7
## + Cautious        4    17788 498285 1172.3
## - InputDevice     4    170658 686731 1174.8
##
## Step: AIC=1155.57
## avg_time ~ Age + AvgSleepTime + CaffeinIntake + UsePrimaryHand +
##      InputDevice + RefreshRate + WiFi + Stress_3 + Shooting_Levels
##
##      Df Sum of Sq    RSS    AIC
## - CaffeinIntake   1    14370 562877 1154.0
## - WiFi           1    18603 567110 1155.0
## <none>                        548507 1155.6
## - RefreshRate     2    49010 597517 1156.9
## - Stress_3        2    50136 598643 1157.1
## + Fatigue_3       2    32434 516073 1157.4
## + Distraction     1    11031 537476 1157.8
## - UsePrimaryHand  1    33482 581989 1158.3
## - Shooting_Levels 2    57337 605844 1158.7
## + Gamer           1     4553 543954 1159.4
## + Game_Frequency  1     3318 545189 1159.7
## + LastNightSleep  1     2410 546097 1159.9
## - AvgSleepTime    1    41446 589953 1160.1
## + HoursAweak      1       952 547555 1160.2
## + Sports_Frequency 1       875 547632 1160.2
## - Age             1    42449 590956 1160.3

```

```

## + logAvgHoursExercise 1      194 548313 1160.4
## + PrimarilyHand      2      7688 540819 1163.5
## + VisualAcuity       4     46810 501697 1163.5
## + Class              3     25855 522652 1163.9
## + System            2      4806 543701 1164.2
## + Device            3     23110 525397 1164.6
## + Noise_Level       2      1086 547421 1165.0
## + Temperature       3      8404 540103 1168.2
## + Cautious          4     23783 524724 1169.3
## - InputDevice       4    164875 713382 1170.0
##
## Step:  AIC=1154.05
## avg_time ~ Age + AvgSleepTime + UsePrimaryHand + InputDevice +
##      RefreshRate + WiFi + Stress_3 + Shooting_Levels
##
##              Df Sum of Sq   RSS   AIC
## - WiFi              1    16704 579581 1153.0
## <none>                562877 1154.0
## - RefreshRate      2    47383 610260 1154.8
## + Distraction      1    17472 545405 1154.8
## - UsePrimaryHand   1    28044 590921 1155.5
## + CaffeinIntake    1    14370 548507 1155.6
## - Stress_3         2    54950 617826 1156.3
## - Shooting_Levels  2    56682 619558 1156.7
## + Gamer            1     8444 554433 1157.0
## + Fatigue_3        2    28501 534376 1157.1
## + Game_Frequency   1     4306 558570 1157.9
## - AvgSleepTime     1    42193 605070 1158.5
## + LastNightSleep   1     1688 561188 1158.5
## + logAvgHoursExercise 1     1178 561698 1158.6
## + HoursAweak       1      909 561968 1158.7
## + Sports_Frequency  1      96 562781 1158.9
## - Age              1    57490 620367 1161.7
## + System           2     6116 556760 1162.4
## + Device           3    26692 536185 1162.4
## + PrimarilyHand    2     6096 556781 1162.4
## + Noise_Level      2     1298 561579 1163.5
## + Class            3    19876 543000 1164.0
## + VisualAcuity     4    39616 523260 1164.1
## + Temperature      3    10323 552554 1166.2
## + Cautious         4    17666 545210 1169.4
## - InputDevice      4   186676 749553 1171.6
##
## Step:  AIC=1152.96
## avg_time ~ Age + AvgSleepTime + UsePrimaryHand + InputDevice +
##      RefreshRate + Stress_3 + Shooting_Levels
##
##              Df Sum of Sq   RSS   AIC
## - RefreshRate      2    38588 618169 1151.5
## <none>                579581 1153.0
## - Stress_3         2    46025 625606 1153.1
## - UsePrimaryHand   1    25145 604725 1153.6
## - Shooting_Levels  2    50150 629731 1154.0
## + WiFi             1    16704 562877 1154.0

```

```

## + Distraction          1      15752 563829 1154.3
## + Gamer                1      12747 566834 1155.0
## + CaffeinIntake        1      12471 567110 1155.0
## + Fatigue_3            2      31821 547760 1155.4
## + Game_Frequency       1       7149 572432 1156.2
## - AvgSleepTime         1      39927 619508 1156.7
## + logAvgHoursExercise  1       1643 577938 1157.5
## + HoursAweak           1       1097 578484 1157.6
## + LastNightSleep       1       552 579029 1157.7
## + Sports_Frequency     1        65 579516 1157.8
## - Age                  1      56084 635665 1160.0
## + System               2       8240 571341 1160.8
## + Device               3      28221 551360 1161.1
## + PrimarilyHand        2       5849 573732 1161.4
## + Noise_Level          2       1237 578344 1162.4
## + Class                3      20137 559444 1163.0
## + VisualAcuity         4      37486 542095 1163.8
## + Temperature          3      12005 567576 1164.8
## + Cautious             4      17982 561599 1168.3
## - InputDevice          4     206012 785593 1172.8
##
## Step:  AIC=1151.55
## avg_time ~ Age + AvgSleepTime + UsePrimaryHand + InputDevice +
##      Stress_3 + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - Stress_3      2     31408 649578 1148.2
## - UsePrimaryHand 1     20059 638228 1150.8
## <none>                      618169 1151.5
## - AvgSleepTime  1     28131 646300 1152.4
## + Distraction   1     18466 599703 1152.5
## + RefreshRate   2     38588 579581 1153.0
## + Gamer         1     12207 605962 1153.8
## + CaffeinIntake 1     11635 606535 1154.0
## + Fatigue_3     2     32187 585983 1154.4
## + WiFi          1       7909 610260 1154.8
## + Game_Frequency 1      6661 611508 1155.0
## + logAvgHoursExercise 1     5245 612925 1155.3
## + HoursAweak    1      3800 614370 1155.6
## - Shooting_Levels 2     70043 688212 1155.7
## + LastNightSleep 1       803 617367 1156.2
## + Sports_Frequency 1        16 618153 1156.4
## + Device        3     41828 576341 1157.1
## - Age           1     58968 677138 1158.5
## + System        2      9841 608328 1159.2
## + PrimarilyHand 2      2122 616047 1160.8
## + Noise_Level   2       333 617836 1161.2
## + Class         3     17642 600527 1162.4
## + VisualAcuity  4     38691 579479 1162.7
## + Temperature   3     16185 601985 1162.7
## + Cautious      4     12811 605358 1168.3
## - InputDevice   4    271562 889731 1179.1
##
## Step:  AIC=1148.23

```



```

## avg_time ~ Age + AvgSleepTime + UsePrimaryHand + InputDevice +
##   Shooting_Levels
##
##           Df Sum of Sq    RSS    AIC
## - UsePrimaryHand      1    11011 660588 1145.5
## + Distraction          1    28950 620627 1147.2
## - AvgSleepTime         1    24124 673701 1148.1
## <none>                  649578 1148.2
## + Gamer                1    16505 633072 1149.8
## + CaffeinIntake         1    15521 634056 1150.0
## + Game_Frequency        1     8352 641225 1151.4
## + Stress_3              2    31408 618169 1151.5
## + HoursAweak            1     4977 644601 1152.1
## + WiFi                  1     3966 645611 1152.3
## + logAvgHoursExercise   1     1322 648256 1152.8
## + Sports_Frequency      1      647 648930 1153.0
## + LastNightSleep        1       20 649558 1153.1
## + RefreshRate           2    23972 625606 1153.1
## + Device                3    45013 604564 1153.5
## + Fatigue_3             2    21485 628092 1153.6
## - Shooting_Levels       2    81570 731148 1153.8
## + System                2    11376 638202 1155.7
## + Class                  3    33514 616064 1156.0
## + PrimarilyHand         2     3633 645945 1157.2
## - Age                   1    75656 725234 1157.6
## + Noise_Level           2     1087 648490 1157.7
## + Temperature           3    12004 637574 1160.4
## + VisualAcuity          4    28589 620989 1161.9
## + Cautious              4    18355 631223 1164.0
## - InputDevice           4   261468 911045 1172.4
##
## Step:  AIC=1145.54
## avg_time ~ Age + AvgSleepTime + InputDevice + Shooting_Levels
##
##           Df Sum of Sq    RSS    AIC
## + Distraction          1    29814 630774 1144.4
## - AvgSleepTime         1    21999 682587 1144.9
## <none>                  660588 1145.5
## + Gamer                1    21720 638869 1146.1
## + CaffeinIntake         1    11642 648946 1148.1
## + UsePrimaryHand        1    11011 649578 1148.2
## + Game_Frequency        1     8848 651740 1148.7
## - Shooting_Levels       2    73292 733880 1149.4
## + HoursAweak            1     4211 656378 1149.6
## + WiFi                  1     3701 656887 1149.7
## + logAvgHoursExercise   1     1219 659369 1150.2
## + Sports_Frequency      1      925 659663 1150.2
## + Fatigue_3             2    25158 635430 1150.2
## + LastNightSleep        1       2 660586 1150.4
## + RefreshRate           2    22553 638035 1150.8
## + Stress_3              2    22360 638228 1150.8
## + Device                3    38732 621856 1152.3
## + Class                  3    35674 624914 1153.0
## + System                2    10872 649716 1153.1

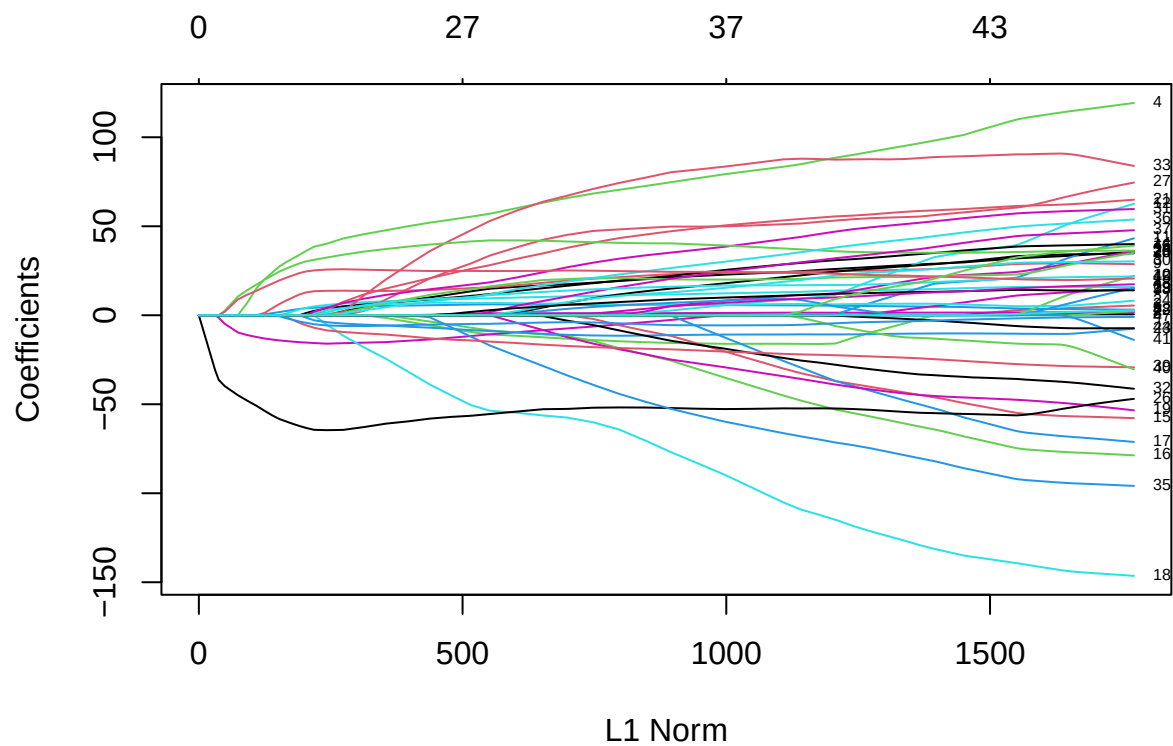
```

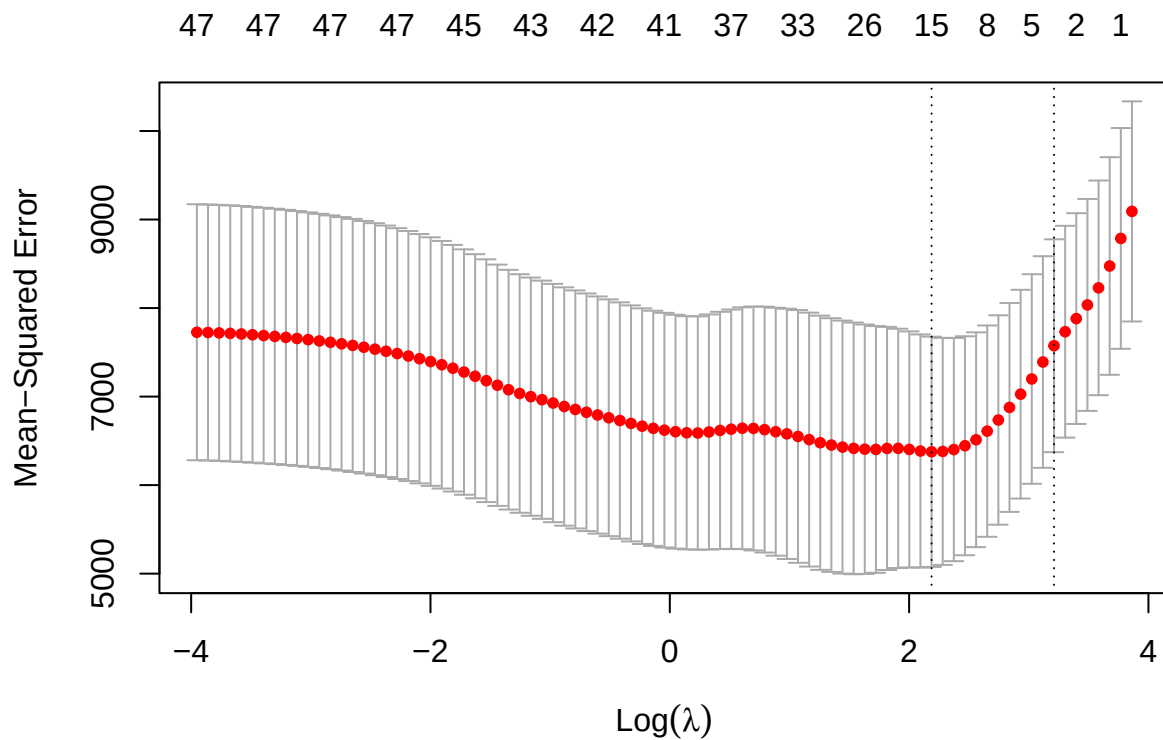
```

## + PrimarilyHand      2      2970 657618 1154.7
## + Noise_Level        2        307 660281 1155.2
## - Age                1      80964 741552 1155.6
## + Temperature        3      13189 647399 1157.5
## + VisualAcuity       4      31287 629301 1158.7
## + Cautious           4      19194 641394 1161.2
## - InputDevice        4     276674 937262 1171.2
##
## Step: AIC=1144.44
## avg_time ~ Age + AvgSleepTime + InputDevice + Shooting_Levels +
##   Distraction
##
##              Df Sum of Sq    RSS    AIC
## <none>                        630774 1144.4
## - AvgSleepTime      1      26881 657655 1145.0
## - Distraction       1      29814 660588 1145.5
## + Gamer             1      16241 614533 1145.9
## + UsePrimaryHand    1      10147 620627 1147.2
## + Game_Frequency    1      10002 620772 1147.2
## - Shooting_Levels   2      66798 697572 1147.7
## + WiFi              1       5756 625018 1148.1
## + CaffeinIntake     1       4188 626586 1148.4
## + logAvgHoursExercise 1       2644 628130 1148.8
## + HoursAweak        1       2039 628735 1148.9
## + Fatigue_3         2      24828 605946 1149.0
## + Sports_Frequency  1       1474 629300 1149.0
## + LastNightSleep    1        333 630441 1149.2
## + RefreshRate       2      21687 609087 1149.7
## + Stress_3          2      14083 616691 1151.2
## + Device            3      31374 599400 1152.4
## + Class             3      30755 600019 1152.6
## + Noise_Level       2       6423 624351 1152.8
## + System            2       6359 624415 1152.8
## + PrimarilyHand     2       3173 627601 1153.5
## - Age              1      75201 705976 1154.1
## + Temperature       3      14039 616735 1156.1
## + VisualAcuity      4      27428 603346 1158.1
## + Cautious          4      14802 615972 1160.8
## - InputDevice       4     247894 878668 1167.8
##
##
## Call:
## lm(formula = avg_time ~ Age + AvgSleepTime + InputDevice + Shooting_Levels +
##   Distraction, data = Survey_cleaned)
##
## Coefficients:
##              (Intercept)                      Age
##              113.638                          15.768
##              AvgSleepTime                      InputDeviceMouse
##              14.512                          -88.056
##              InputDeviceKeyboard                InputDeviceTrackpad
##              56.524                          8.402
##              InputDeviceTouch screen  Shooting_LevelsIntermediate Levels
##              -28.410                          22.652

```

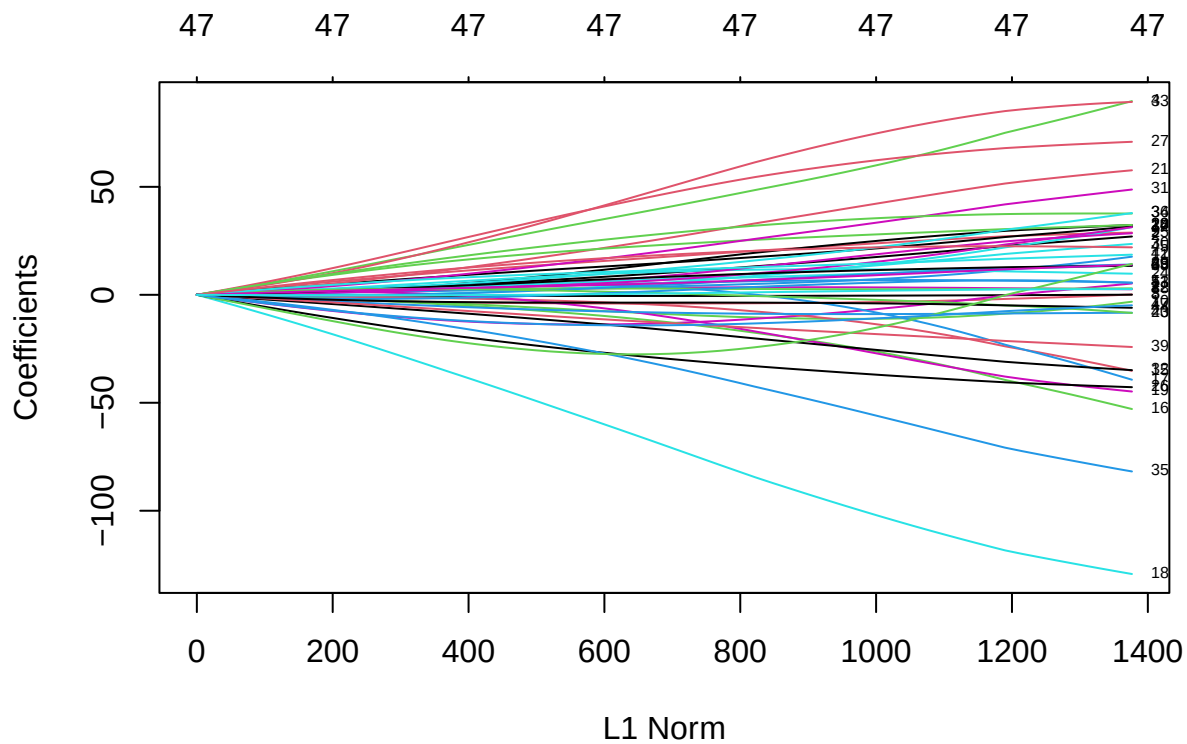
##	Shooting_LevelsLow Levels	DistractionYes
##	58.928	37.854

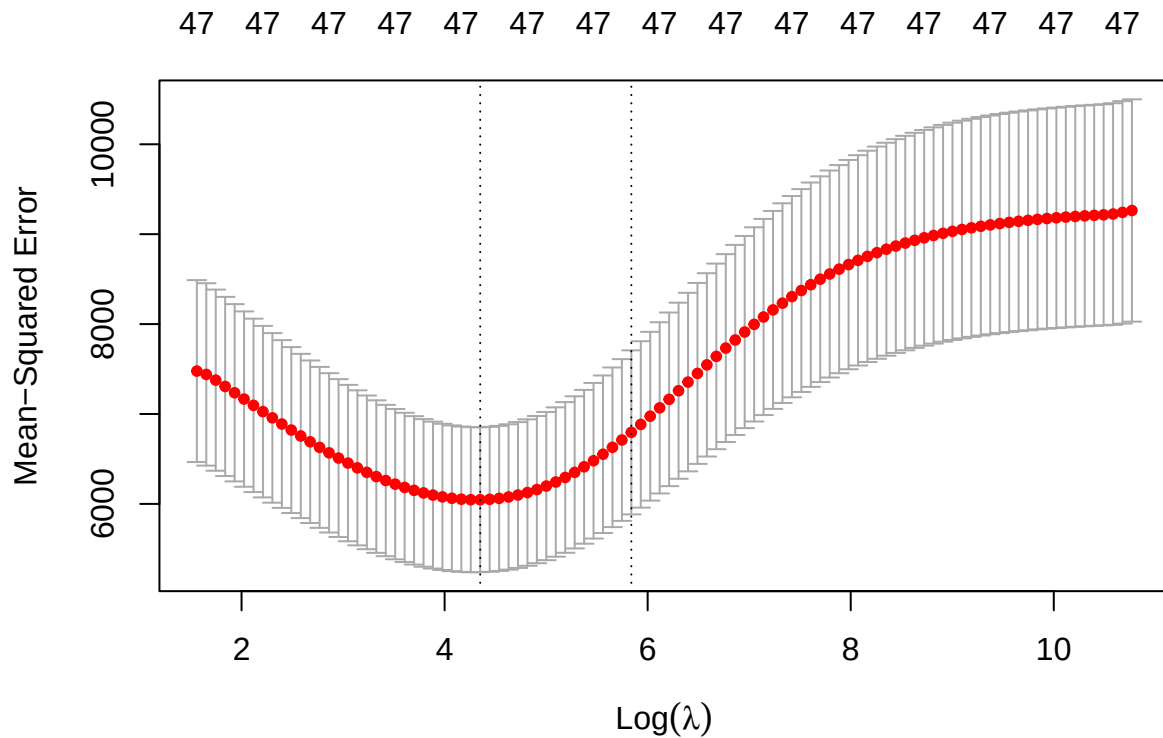




```
## 49 x 1 sparse Matrix of class "dgCMatrix"
##
## (Intercept) 434.637178
## (Intercept) .
## ClassJunior .
## ClassSenior .
## ClassGraduate student 38.495659
## Age 4.207794
## AvgSleepTime 4.959268
## LastNightSleep .
## HoursAweek .
## DistractionYes 13.502323
## TemperatureWarm .
## TemperatureNeutral .
## TemperatureCold .
## GamerYes -15.617365
## CaffeinIntakeYes 2.454916
## VisualAcuityExcellent .
## VisualAcuityGood .
## VisualAcuityPoor .
## VisualAcuityVery Poor .
## PrimarilyHandLeft hand .
## PrimarilyHandAmbidextrous (both hands equally) .
## UsePrimaryHandNo .
## CautiousVery cautious .
## CautiousModerately cautious -2.849909
```

## CautiousSlightly cautious	1.386369
## CautiousNot cautious at all	.
## InputDeviceMouse	-64.358406
## InputDeviceKeyboard	.
## InputDeviceTrackpad	0.167877
## InputDeviceTouch screen	.
## RefreshRateMid-Range	.
## RefreshRateStandard	2.010429
## DeviceiPad(Tablet	.
## DeviceiPhone(Smartphone	.
## DeviceLaptop	31.126876
## WiFiStable	.
## Fatigue_3Moderately Fatigue	.
## Fatigue_3Low Fatigue	.
## Stress_3Moderately Stress	.
## Stress_3Low Stress	-7.020384
## Noise_LevelModerately Noise	.
## Noise_LevelLow Noise	.
## Game_FrequencyLow Frequency	.
## Sports_FrequencyLow Frequency	.
## Shooting_LevelsIntermediate Levels	.
## Shooting_LevelsLow Levels	25.202538
## SystemLinux	.
## SystemWindows	-5.059475
## logAvgHoursExercise	.





```
## 49 x 1 sparse Matrix of class "dgCMatrix"
##
## (Intercept) 368.5377698
## (Intercept) .
## ClassJunior 9.2179213
## ClassSenior -4.0525003
## ClassGraduate student 37.7032688
## Age 5.1690070
## AvgSleepTime 6.5175895
## LastNightSleep 3.1701663
## HoursAweek -0.5404607
## DistractionYes 16.6195321
## TemperatureWarm -8.2325000
## TemperatureNeutral 3.5619055
## TemperatureCold 4.6298805
## GamerYes -13.5314439
## CaffeinIntakeYes 13.9420796
## VisualAcuityExcellent -4.0959600
## VisualAcuityGood -11.0979381
## VisualAcuityPoor 4.1943588
## VisualAcuityVery Poor -64.8507779
## PrimarilyHandLeft hand -8.0711916
## PrimarilyHandAmbidextrous (both hands equally) 13.1496551
## UsePrimaryHandNo 23.8720903
## CautiousVery cautious 2.9004930
## CautiousModerately cautious -7.9892714
```

```

## CautiousSlightly cautious          10.2090207
## CautiousNot cautious at all        7.5536393
## InputDeviceMouse                   -28.1947413
## InputDeviceKeyboard                43.8645305
## InputDeviceTrackpad                22.3201868
## InputDeviceTouch screen            3.9222785
## RefreshRateMid-Range               6.6727554
## RefreshRateStandard                18.5185918
## DeviceiPad(Tablet                  -14.9492712
## DeviceiPhone(Smartphone            45.4042015
## DeviceLaptop                       26.9626098
## WiFiStable                         -29.9684480
## Fatigue_3Moderately Fatigue        10.8075295
## Fatigue_3Low Fatigue               6.1120911
## Stress_3Moderately Stress           7.7363361
## Stress_3Low Stress                 -12.1926836
## Noise_LevelModerately Noise         1.0909481
## Noise_LevelLow Noise               1.2814921
## Game_FrequencyLow Frequency        11.0270908
## Sports_FrequencyLow Frequency       4.9663124
## Shooting_LevelsIntermediate Levels -3.6278837
## Shooting_LevelsLow Levels          17.7413953
## SystemLinux                        -27.5803947
## SystemWindows                      -14.1677818
## logAvgHoursExercise                0.4618736

##
## Call:
## lm(formula = avg_time ~ Class + AvgSleepTime + CaffeinIntake +
##     VisualAcuity + PrimarilyHand + UsePrimaryHand + InputDevice +
##     RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Game_Frequency,
##     data = Survey_cleaned)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -138.894  -31.697   -6.822   33.277  192.712
##
## Coefficients:
##                                     Estimate Std. Error t value
## (Intercept)                       425.317    105.952    4.014
## ClassJunior                        38.217     15.069    2.536
## ClassSenior                        13.981     18.142    0.771
## ClassGraduate student             130.035     28.253    4.602
## AvgSleepTime                       15.003      6.128    2.448
## CaffeinIntakeYes                   39.404     14.849    2.654
## VisualAcuityExcellent              -69.883     25.778   -2.711
## VisualAcuityGood                  -82.245     26.164   -3.143
## VisualAcuityPoor                  -80.451     42.878   -1.876
## VisualAcuityVery Poor             -117.412     75.133   -1.563
## PrimarilyHandLeft hand             -41.362     35.280   -1.172
## PrimarilyHandAmbidextrous (both hands equally) 32.110     36.761    0.873
## UsePrimaryHandNo                   71.433     33.181    2.153
## InputDeviceMouse                   -31.564     68.331   -0.462
## InputDeviceKeyboard                68.597     85.193    0.805

```

```

## InputDeviceTrackpad          45.508      67.390      0.675
## InputDeviceTouch screen      -3.979      75.914     -0.052
## RefreshRateMid-Range         28.670      19.571      1.465
## RefreshRateStandard          55.821      18.704      2.984
## DeviceiPad(Tablet            -11.942      53.720     -0.222
## DeviceiPhone(Smartphone      140.629      63.237      2.224
## DeviceLaptop                 38.479      21.151      1.819
## WiFiStable                   -65.223      49.872     -1.308
## Fatigue_3Moderately Fatigue  62.065      20.502      3.027
## Fatigue_3Low Fatigue         54.603      20.072      2.720
## Stress_3Moderately Stress     7.351      17.207      0.427
## Stress_3Low Stress           -35.489      18.291     -1.940
## Game_FrequencyLow Frequency  26.502      12.463      2.127
##                               Pr(>|t|)
## (Intercept)                  0.000115 ***
## ClassJunior                   0.012744 *
## ClassSenior                   0.442730
## ClassGraduate student        1.22e-05 ***
## AvgSleepTime                  0.016081 *
## CaffeinIntakeYes              0.009250 **
## VisualAcuityExcellent         0.007885 **
## VisualAcuityGood              0.002193 **
## VisualAcuityPoor              0.063504 .
## VisualAcuityVery Poor        0.121248
## PrimarilyHandLeft hand       0.243804
## PrimarilyHandAmbidextrous (both hands equally) 0.384468
## UsePrimaryHandNo             0.033711 *
## InputDeviceMouse              0.645123
## InputDeviceKeyboard           0.422595
## InputDeviceTrackpad           0.501031
## InputDeviceTouch screen      0.958303
## RefreshRateMid-Range         0.146044
## RefreshRateStandard           0.003564 **
## DeviceiPad(Tablet            0.824536
## DeviceiPhone(Smartphone      0.028387 *
## DeviceLaptop                  0.071840 .
## WiFiStable                    0.193907
## Fatigue_3Moderately Fatigue  0.003131 **
## Fatigue_3Low Fatigue         0.007680 **
## Stress_3Moderately Stress     0.670124
## Stress_3Low Stress            0.055140 .
## Game_FrequencyLow Frequency  0.035894 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 64.09 on 101 degrees of freedom
## Multiple R-squared:  0.6454, Adjusted R-squared:  0.5506
## F-statistic: 6.808 on 27 and 101 DF,  p-value: 4.733e-13

##                               GVIF Df GVIF^(1/(2*Df))
## Class                1.882939  3      1.111235
## AvgSleepTime         1.242119  1      1.114504
## CaffeinIntake         1.291534  1      1.136457
## VisualAcuity          1.918683  4      1.084864

```



```
## PrimarilyHand 2.157614 2 1.211974
## UsePrimaryHand 1.774313 1 1.332033
## InputDevice 6.483538 4 1.263213
## RefreshRate 2.266885 2 1.227036
## Device 6.241066 3 1.356885
## WiFi 1.192216 1 1.091886
## Fatigue_3 1.749256 2 1.150041
## Stress_3 1.897225 2 1.173626
## Game_Frequency 1.198209 1 1.094627
```

```
## Analysis of Variance Table
##
```

```
## Response: avg_time
```

```
##          Df Sum Sq Mean Sq F value    Pr(>F)
## Class          3  70764    23588   4.6315 0.0047165 **
## AvgSleepTime    1  51358    51358  10.0842 0.0020692 **
## CaffeinIntake    1  44474    44474   8.7324 0.0040213 **
## VisualAcuity     4 112552    28138   5.5249 0.0005194 ***
## UsePrimaryHand    1   8500     8500   1.6689 0.1998229
## RefreshRate      2 133463    66731  13.1028 1.061e-05 ***
## Device           3  90243    30081   5.9064 0.0010229 **
## WiFi             1  19165    19165   3.7631 0.0556363 .
## Fatigue_3        2  29443    14722   2.8906 0.0608947 .
## Stress_3         2  33627    16814   3.3014 0.0414977 *
## Game_Frequency   1  31181    31181   6.1224 0.0152893 *
## Class:AvgSleepTime 3   4805     1602   0.3145 0.8148506
## AvgSleepTime:CaffeinIntake 1   2820     2820   0.5538 0.4587899
## AvgSleepTime:VisualAcuity 3   4600     1533   0.3011 0.8245341
## AvgSleepTime:PrimarilyHand 2  19859     9929   1.9497 0.1484902
## AvgSleepTime:UsePrimaryHand 1   4988     4988   0.9794 0.3250906
## AvgSleepTime:RefreshRate 2  21825    10912   2.1426 0.1234963
## AvgSleepTime:Device 2  20285    10143   1.9915 0.1426606
## AvgSleepTime:WiFi 1    162      162   0.0318 0.8588795
## AvgSleepTime:Fatigue_3 2  13297     6648   1.3054 0.2763119
## AvgSleepTime:Stress_3 2   8767     4383   0.8607 0.4264506
## AvgSleepTime:Game_Frequency 1    694      694   0.1363 0.7129253
## Residuals       87 443084     5093
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## Analysis of Variance Table
##
```

```
## Response: avg_time
```

```
##          Df Sum Sq Mean Sq F value    Pr(>F)
## Class          3  70764    23588   4.6294 0.0043877 **
## AvgSleepTime    1  51358    51358  10.0797 0.0019583 **
## CaffeinIntake    1  44474    44474   8.7285 0.0038528 **
## VisualAcuity     4 112552    28138   5.5225 0.0004426 ***
## UsePrimaryHand    1   8500     8500   1.6682 0.1992846
## RefreshRate      2 133463    66731  13.0969 8.168e-06 ***
## Device           3  90243    30081   5.9038 0.0009104 ***
## WiFi             1  19165    19165   3.7614 0.0550824 .
## Fatigue_3        2  29443    14722   2.8893 0.0599655 .
## Stress_3         2  33627    16814   3.2999 0.0406746 *
```

```

## Game_Frequency    1  31181    31181  6.1196 0.0149411 *
## Residuals         107 545186     5095
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## Analysis of Variance Table
##
## Response: avg_time
##              Df Sum Sq Mean Sq F value    Pr(>F)
## Class          3  70764   23588   4.4860 0.0052020 **
## AvgSleepTime    1  51358   51358   9.7673 0.0022721 **
## CaffeinIntake    1  44474   44474   8.4581 0.0043975 **
## VisualAcuity     4 112552   28138   5.3513 0.0005638 ***
## RefreshRate      2 124040   62020 11.7950 2.286e-05 ***
## Device           3  82181   27394   5.2098 0.0021120 **
## Fatigue_3        2  39237   19619   3.7311 0.0270521 *
## Game_Frequency   1  44518   44518   8.4664 0.0043787 **
## UsePrimaryHand    1  22436   22436   4.2668 0.0412128 *
## Residuals       110 578395     5258
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##              GVIF Df GVIF^(1/(2*Df))
## Age             1.083396  1      1.040863
## AvgSleepTime     1.064404  1      1.031699
## InputDevice       1.183969  4      1.021333
## Shooting_Levels  1.095841  2      1.023144
## Distraction       1.053618  1      1.026459

##
## Call:
## lm(formula = avg_time ~ Age + AvgSleepTime + InputDevice + Shooting_Levels +
##     Distraction, data = Survey_cleaned)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -128.316  -49.446   -2.732   51.405  201.703
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    113.638    124.733   0.911 0.364110
## Age             15.768      4.186   3.767 0.000259 ***
## AvgSleepTime    14.512      6.444   2.252 0.026159 *
## InputDeviceMouse -88.056    74.274  -1.186 0.238156
## InputDeviceKeyboard  56.524    89.861   0.629 0.530541
## InputDeviceTrackpad   8.402    73.690   0.114 0.909417
## InputDeviceTouch screen -28.410    76.999  -0.369 0.712813
## Shooting_LevelsIntermediate Levels  22.652    20.549   1.102 0.272535
## Shooting_LevelsLow Levels    58.928    18.504   3.185 0.001851 **
## DistractionYes    37.854    15.961   2.372 0.019315 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##

```

```
## Residual standard error: 72.81 on 119 degrees of freedom
## Multiple R-squared:  0.4609, Adjusted R-squared:  0.4201
## F-statistic: 11.3 on 9 and 119 DF,  p-value: 1.231e-12
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: avg_time
```

```
##
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
## Age	1	37863	37863	5.9245	0.0164430 *
## AvgSleepTime	1	60340	60340	9.4417	0.0026391 **
## Shooting_Levels	2	134489	67245	10.5220	6.278e-05 ***
## Distraction	1	58594	58594	9.1684	0.0030297 **
## Age:Shooting_Levels	2	120026	60013	9.3905	0.0001651 ***
## Age:Distraction	1	1556	1556	0.2435	0.6225874
## AvgSleepTime:Shooting_Levels	2	9066	4533	0.7093	0.4941076
## AvgSleepTime:Distraction	1	288	288	0.0451	0.8321666
## Residuals	117	747731	6391		

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: avg_time
```

```
##
```

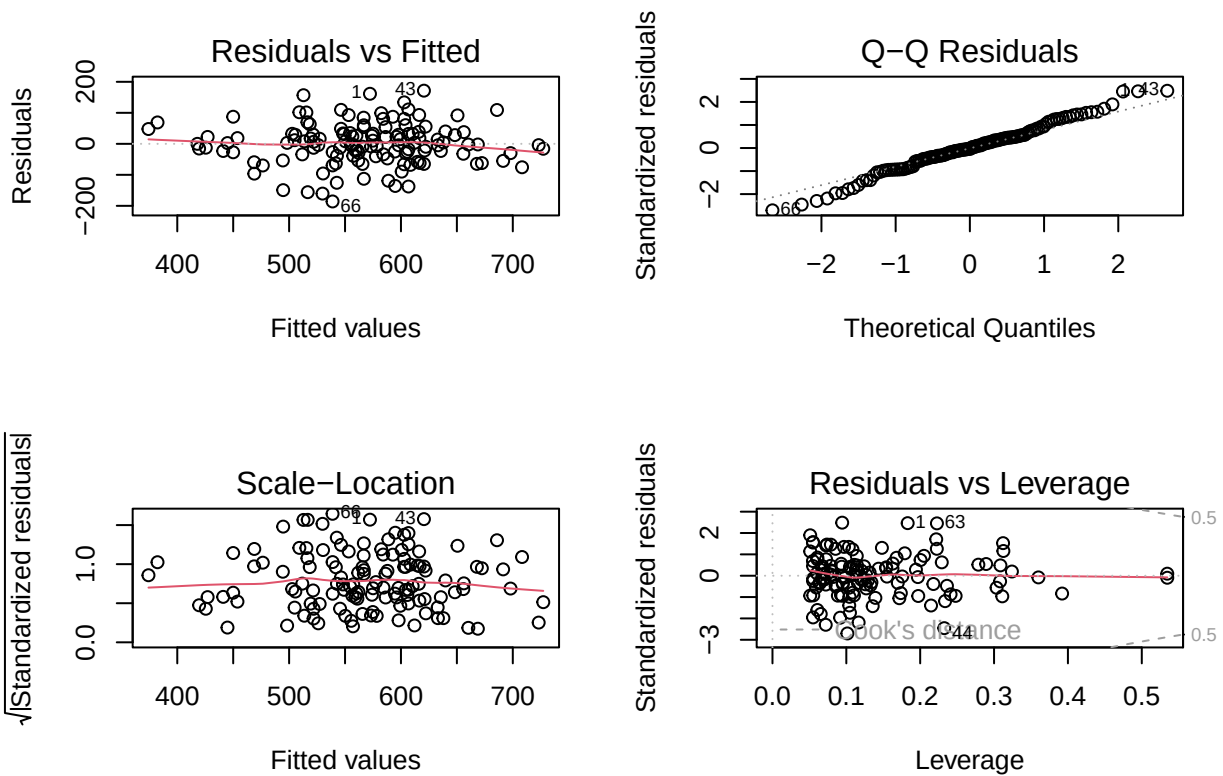
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
## Age	1	37863	37863	6.0390	0.0154107 *
## AvgSleepTime	1	60340	60340	9.6240	0.0023906 **
## Shooting_Levels	2	134489	67245	10.7252	5.151e-05 ***
## Distraction	1	58594	58594	9.3455	0.0027516 **
## Age:Shooting_Levels	2	120026	60013	9.5718	0.0001383 ***
## Residuals	121	758642	6270		

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## Warning: not plotting observations with leverage one:
```

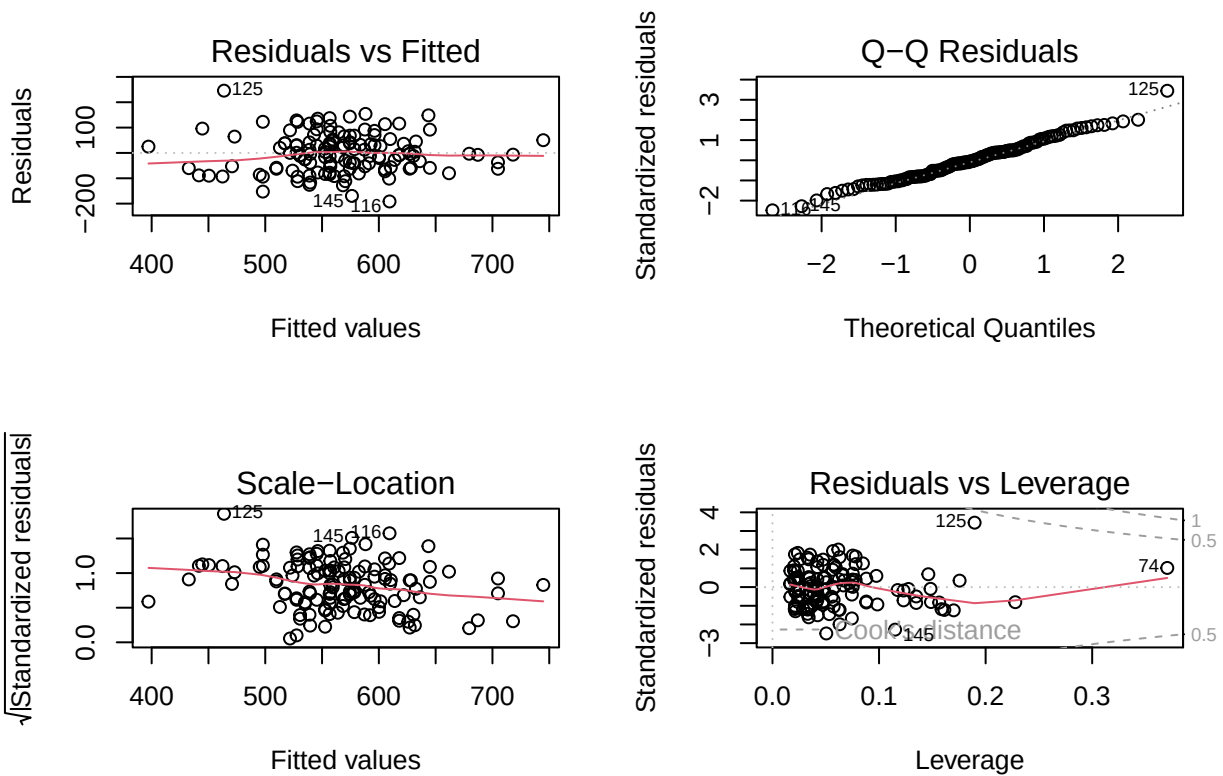
```
## 14
```



```
##
## studentized Breusch-Pagan test
##
## data: AICmod4
## BP = 26.426, df = 18, p-value = 0.0904

##
## Shapiro-Wilk normality test
##
## data: residuals(AICmod4)
## W = 0.98837, p-value = 0.3476

##
## Durbin-Watson test
##
## data: AICmod4
## DW = 2.1674, p-value = 0.8238
## alternative hypothesis: true autocorrelation is greater than 0
```



```
##
## studentized Breusch-Pagan test
##
## data: BICmod3
## BP = 18.42, df = 7, p-value = 0.01021

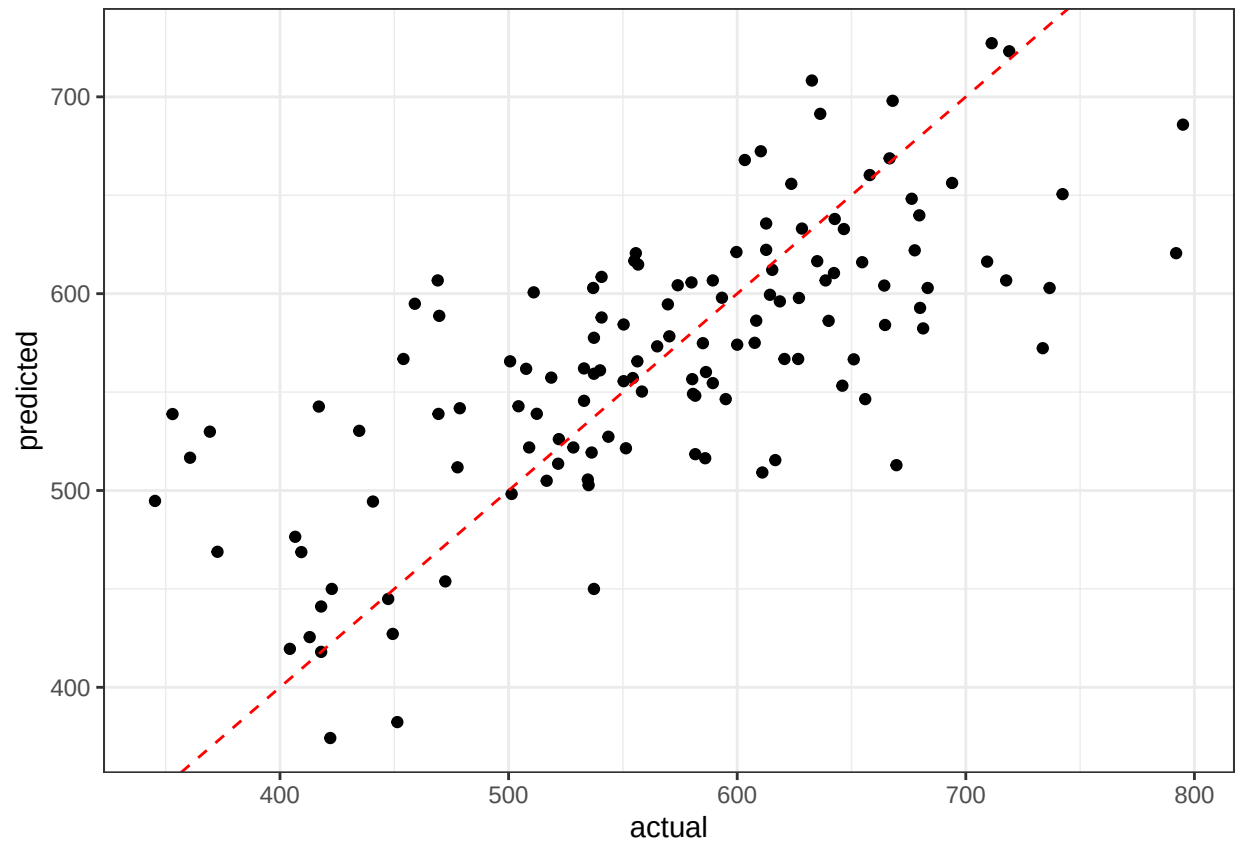
##
## Shapiro-Wilk normality test
##
## data: residuals(BICmod3)
## W = 0.99066, p-value = 0.5411

##
## Durbin-Watson test
##
## data: BICmod3
## DW = 1.649, p-value = 0.02156
## alternative hypothesis: true autocorrelation is greater than 0

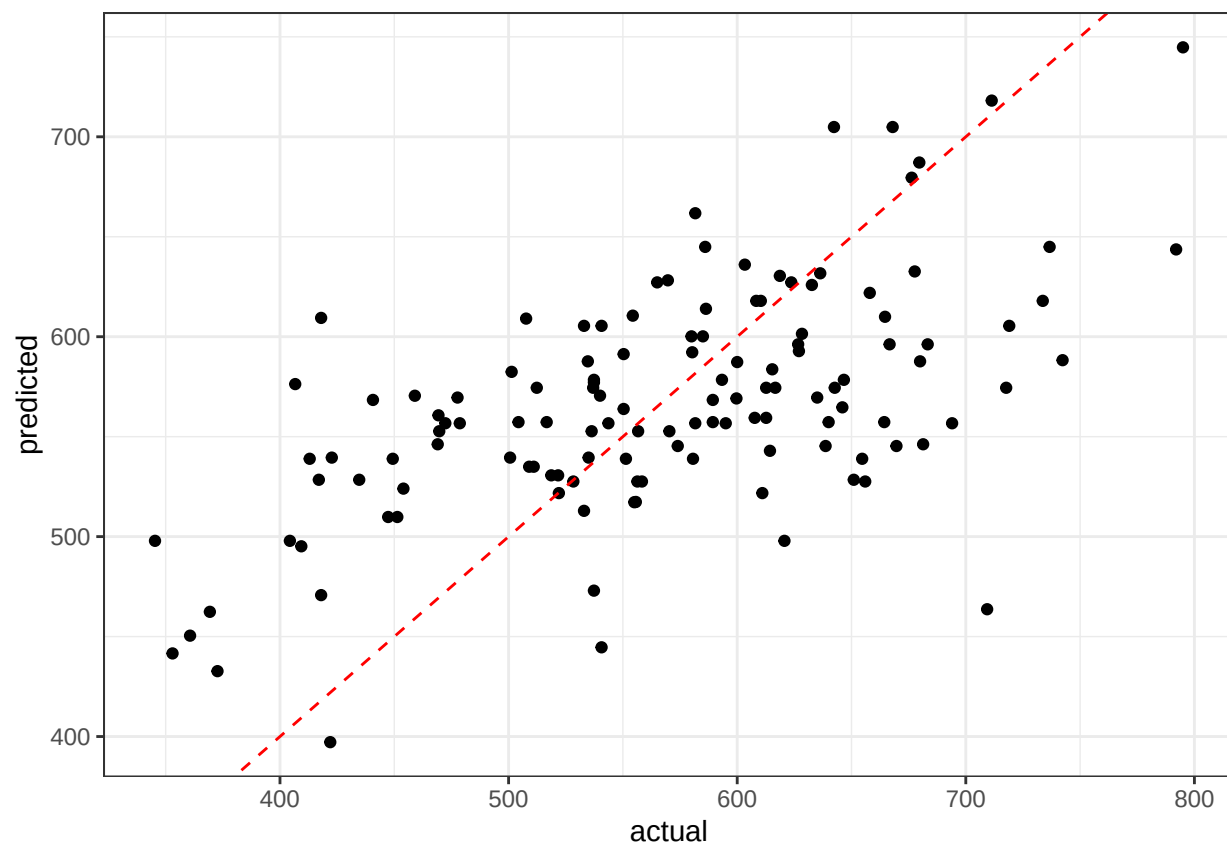
##
## studentized Breusch-Pagan test
##
## data: BICmod3_weighted
## BP = 9.2754, df = 7, p-value = 0.2335
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: residuals(BICmod3_weighted)  
## W = 0.99034, p-value = 0.511
```

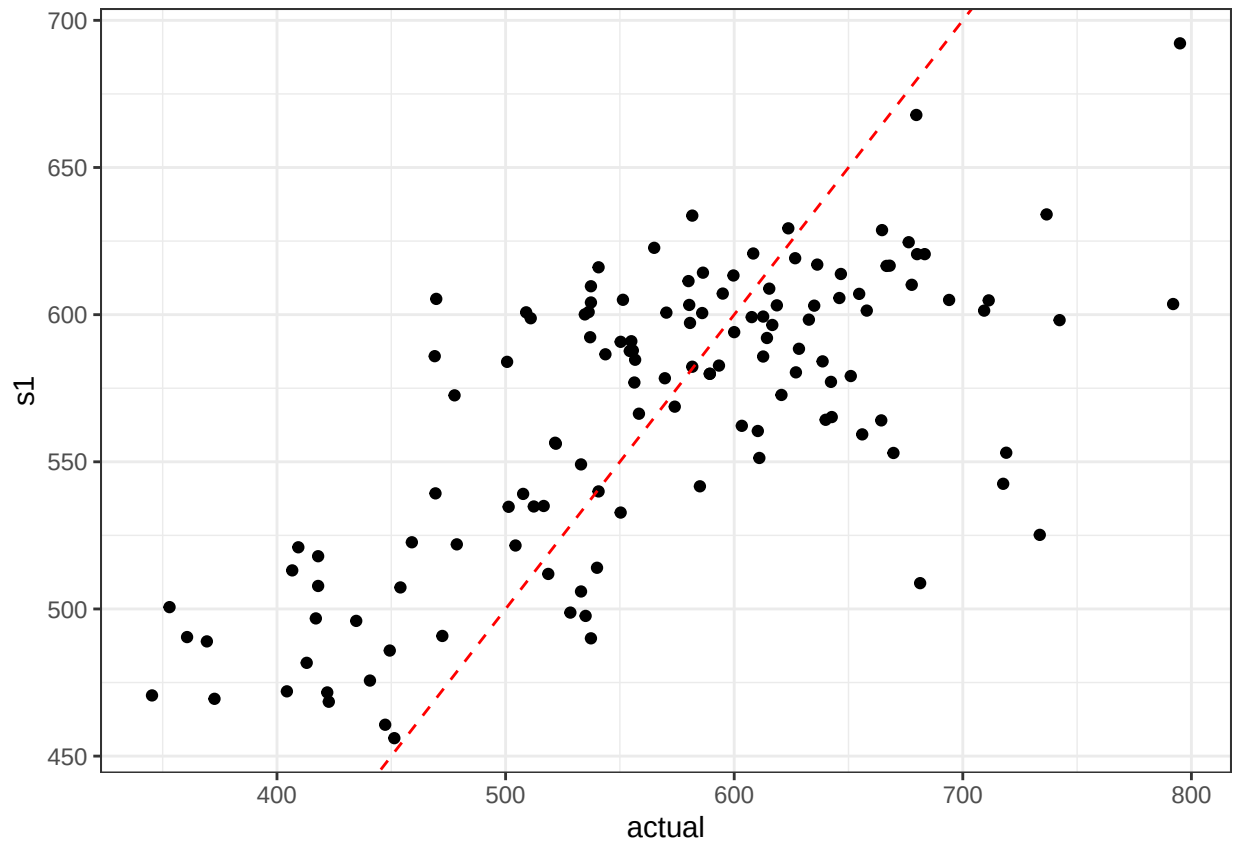
```
## [1] 66.96031
```



```
## [1] 76.78642
```



```
## [1] 70.01466
```



```
## [1] 62.5103
```